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Governing Digital Platform Ecosystems in Enterprise Digital Transformation: A Scoping Review and Dialectical Systems Framework

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Data availability

Data will be made available on request.

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Governing digital platform ecosystems in enterprise digital transformation: a scoping review and dialectical systems framework

Abstract:

This scoping review examines how digital platforms shape enterprise digital transformation by mapping the structure, mechanisms, and governance logic of a fragmented interdisciplinary literature. Following PRISMA-ScR guidelines, we searched Web of Science Core Collection (inception-18 June 2025) and Scopus (inception-18 June 2025) for peer-reviewed English-language journal articles and book chapters. After screening and eligibility assessment, 152 studies were included. No effect measures, meta-analysis, or formal study-level risk of bias assessment were conducted because the review focused on evidence mapping and conceptual synthesis. Bibliometric mapping and qualitative content analysis were combined to identify the intellectual structure of the field and synthesize how digital platforms enable, constrain, and govern enterprise transformation. Based on this synthesis, we developed a dialectical systems framework organized around three interconnected dimensions: concept, method, and value. The findings show that digital platform-enabled enterprise digital transformation is not simply a linear process of platform adoption, but an evolving system shaped by recurring tensions, including openness versus control, efficiency versus resilience, and data sharing versus privacy. This study explains how these tensions are coordinated across platform design, empirical representation, and stakeholder value claims, providing a basis for future research on platform governance and transformation dynamics.

Keywords: Digital platform; Enterprise digital transformation; Scoping review; Dialectical systems framework; Platform governance; Conflict-coordination dynamics

1. Introduction

Enterprises are accelerating digital transformation to maintain competitiveness and organizational adaptability (Ji et al., 2025). Digital platforms have become a central organizational and technological infrastructure in this process. Built on cloud computing, data infrastructure, and artificial intelligence, they connect users, services, and resources through shared digital architectures (Zhao and Zha, 2024). They support real-time information exchange, data-driven coordination, production efficiency, supply chain integration, and service-based business models across consumer and industrial contexts (Baronian, 2020; Ben Arfi and Hikkerova, 2021; Hasselwander, 2024; Sauvagerd et al., 2024).

The role of digital platforms in enterprise digital transformation extends beyond technological upgrading. Platforms reshape value chains, reconfigure firm boundaries, and alter the logic of value creation. Prior studies have identified several capability domains that support this transformation, including technical architecture, data integration, business expansion, and intelligent applications (Cui and Liu, 2012; Ferreira et al., 2021; Liu et al., 2024; Ruutu et al., 2017). Through resource integration and value co-creation, firms can move from process improvement to broader system-level value creation (Jesus et al., 2024; Ruggieri et al., 2018; Wang, 2025). Digital platforms also connect digital technologies with the real economy

and generate wider economic and social effects (Dabbous et al., 2023; Huo et al., 2024).

Despite growing interest in digital platform-enabled enterprise digital transformation (DP-EDT), the literature remains fragmented (Okano et al., 2021). Existing studies often examine technological, organizational, or strategic factors through separate analytical lenses. This fragmentation limits understanding of how platform design, governance mechanisms, transformation pathways, and stakeholder value claims interact across levels and change over time (Mahroof et al., 2025; Pan and Ning, 2014; Yang et al., 2025). The use of established frameworks also reflects this fragmentation. The TOE framework explains adoption conditions, dynamic capabilities theory explains capability renewal and adaptation, and ecosystem governance models explain inter-organizational coordination. These perspectives provide important insights into DP-EDT, while their connections remain underdeveloped. A more integrated framework is needed to explain how cross-level interactions, persistent tensions, and multi-actor coordination jointly shape platform-enabled transformation.

This study adopts a dialectical systems perspective to address this gap. This perspective views DP-EDT as a process shaped by interdependent and competing forces. Recurrent tensions include openness and control, efficiency and resilience, data sharing and privacy, and short-term value capture and long-term ecosystem viability (Schmidt et al., 2019; Wang et al., 2024). Building on prior uses of dialectical thinking in management research (Pan and Ning, 2014; Schmidt et al., 2019), this study develops an analytical framework that brings these tensions into a single structure. The framework links three dimensions: concept, method, and value. The concept dimension captures the substantive components of transformation. The method dimension captures how transformation is observed, measured, and interpreted in existing research. The value dimension captures how stakeholder interests are created, constrained, redistributed, and coordinated.

To operationalize this perspective, we conducted a scoping review of 152 peer-reviewed studies following the PRISMA-ScR guidelines. A scoping review is suitable for DP-EDT because the literature is conceptually fragmented, methodologically diverse, and distributed across multiple disciplines. The review maps the scope, structure, concepts, methods, and stakeholder value configurations of the field. Bibliometric mapping was used to identify thematic clusters and structural patterns, and qualitative content analysis was used to synthesize mechanisms, governance arrangements, and coordination logic. The analysis was organized around three dimensions. The concept dimension covers drivers, platform types, transformation pathways, governance mechanisms, and outcomes. The method dimension summarizes research designs, data sources, analytical methods, and evaluation practices. The value dimension maps stakeholder configurations and conflict-coordination dynamics among firms, customers, service providers, platform operators, regulators, and researchers.

Accordingly, this study addresses three research questions:

RQ1. What are the core components of DP-EDT and how are they interrelated?

RQ2. How do these components interact through conflict and coordination across the conceptual, methodological, and value dimensions?

RQ3. How can a dialectical system framework integrate cross-level transformation processes and clarify their broader economic and social implications?

This study makes three contributions. First, it synthesizes dispersed research on DP-EDT and clarifies the field's core components, methodological patterns, and stakeholder value configurations. Second, it develops a dialectical systems framework that connects concept, method, and value as interdependent dimensions of platform-enabled transformation. Third, it advances a conflict–coordination view of platform governance by explaining how platform design, empirical representation, and stakeholder value claims jointly shape transformation outcomes.

The remainder of this paper is organized as follows. Section 2 explains the research design and screening procedure. Section 3 presents the intellectual structure of the field. Section 4 analyzes DP-EDT across the concept, method, and value dimensions. Section 5 integrates these dimensions and discusses the dialectical relationships among platform design, research methods, and stakeholder value claims. Section 6 concludes the study.

2. Research methods and procedures

2.1. Research methods

This study adopted a scoping review design and was reported in accordance with the PRISMA-ScR checklist. A scoping review was appropriate because research on DP-EDT is conceptually fragmented, methodologically diverse, and distributed across several disciplines. The review focused on the scope, structure, conceptual boundaries, methodological patterns, and stakeholder value configurations of the literature. The review protocol and working materials were deposited in an OSF project repository on 5 June 2025 (osf.io/vce3x). The repository documents the review questions, search strategy, eligibility criteria, planned bibliometric mapping, qualitative coding procedures, and data extraction structure. The PRISMA flow diagram is presented in Fig. 1, and the completed PRISMA-ScR checklist is provided in Supplementary Material A.

The review design focused on evidence mapping and conceptual synthesis. Effect measures, effect size estimation, and quantitative intervention synthesis were outside the scope of the review. No meta-analysis was conducted. No formal study-level risk of bias assessment or methodological quality appraisal was performed because the included studies varied widely in research design, sector, data source, and outcome definition. The limitations of the included evidence were summarized at the review level, with attention to study design, outcome scope, sectoral coverage, and reporting detail.

The review followed a two-stage analytical procedure. First, bibliometric analysis was used to map publication trends, thematic clusters, and intellectual structures in DP-EDT research. This stage identified the structural fragmentation of the field and the dominant research strands. Second, qualitative content analysis was used to interpret the conceptual relationships, governance mechanisms, stakeholder configurations, and conflict–coordination dynamics embedded in these strands. This sequential mapping-and-synthesis approach supported the descriptive and interpretive objectives of the scoping review.

The key data domains extracted from the included studies were organized around three analytical dimensions. The concept dimension covered transformation drivers, platform types, transformation pathways, governance mechanisms, and effects. The method dimension

covered research design, data source, analytical method, and measurement approach. The value dimension covered stakeholder types, value claims, conflicts, and coordination mechanisms. Relevant results were extracted when they directly informed at least one of these domains. When a study reported multiple platform types, sectors, or stakeholder groups, multiple codes were assigned. When information was absent or unclear, it was coded as “not reported” and discussed during coding meetings where necessary.

Building on this coding structure, we developed a dialectical systems framework organized around concept, method, and value. Bibliometric mapping identified the structural patterns of the field, and qualitative content analysis reconstructed how tensions and coordination mechanisms operate across levels. The integration of these approaches enabled a transparent synthesis of DP-EDT research and provided the basis for explaining platform-enabled transformation as a process shaped by interdependence, tension, and dynamic coordination.

2.1.1. Bibliometric analysis

Bibliometric analysis was used to map the intellectual structure and thematic development of DP-EDT research (Broadus, 1987). This analysis provided a descriptive overview of how the field has developed across publication outlets, time periods, geographical contexts, and recurring keywords.

Specifically, we examined four bibliometric indicators: journal distribution, annual publication trends, regional distribution of contributing studies, and keyword co-occurrence networks. Journal distribution was used to assess the disciplinary dispersion of the field. Annual publication trends were used to trace the growth of DP-EDT research over time. Regional distribution was used to identify the main geographical contexts represented in the literature. Keyword co-occurrence analysis was used to identify dominant themes and the relationships among them. Together, these indicators provided the structural basis for the subsequent qualitative synthesis and the development of the dialectical systems framework (Li et al., 2025a).

For bibliometric mapping and subsequent coding, we extracted publication year, journal or source title, country or region, author keywords, Keywords Plus where available, sector, platform type, research design, data source, analytical method, core concepts or measures, stakeholder groups, and funding or conflict-of-interest information when reported. Missing information was coded as “not reported” and was not imputed. Ambiguous classifications were discussed by the coding team and recorded in the coding log.

2.1.2. Content analysis

Content analysis was used to interpret the conceptual structures, governance mechanisms, stakeholder relationships, and conflict–coordination patterns in the DP-EDT literature (Li et al., 2025a; Wang and Guo, 2022). It extended the bibliometric analysis by examining how platform-enabled transformation was explained, operationalized, and evaluated across studies (Suraj et al., 2024).

Bibliometric clustering informed the initial coding structure. All 152 included studies contributed to the bibliometric mapping and were also considered in the qualitative content analysis. The depth of qualitative interpretation varied according to each article’s relevance to the review questions and the richness of its reported evidence. Core studies within each

bibliometric cluster were identified using two criteria: structural relevance in the keyword or citation network and substantive contribution to at least one analytical dimension of the review. These core studies helped refine category definitions and illustrate recurring mechanisms. Full-dataset coding remained the basis of the synthesis.

The qualitative analysis followed a structured multi-stage coding procedure. First, a preliminary coding framework was developed from the research questions, bibliometric clusters, and three analytical dimensions of concept, method, and value. Pilot coding was then conducted on a randomly selected subset of approximately 20% of the included studies to refine category definitions. Two coders independently applied the coding scheme to this subset. Inter-coder reliability was assessed using Cohen's kappa, which yielded an overall value of 0.82 and indicated substantial agreement. Discrepancies mainly concerned boundary cases between overlapping categories.

Coding disagreements were resolved through discussion, with a third senior researcher adjudicating unresolved cases. The coding framework was iteratively refined, and coding decisions were recorded in a coding log. After reliability was established, the remaining studies were coded using the finalized framework.

No formal study-level risk of bias assessment or methodological quality appraisal was conducted. This choice reflected the scoping purpose of the review and the heterogeneity of the included studies in research design, sector, data source, and outcome definition. Pooled effect estimation and intervention-effect claims were outside the scope of the review. The limitations of the mapped evidence were summarized at the review level. No formal sensitivity analysis was performed because the review did not include statistical synthesis. Conceptual heterogeneity was examined qualitatively across platform type, sector, organizational context, stakeholder configuration, research method, and transformation outcome.

During full-text assessment and coding, studies were excluded when digital platforms were only marginally referenced or when the full text lacked sufficient analytical detail for extraction. "Marginal reference" referred to studies in which digital platforms appeared only as background context, keywords, or a general technology label, without analysis of platform-enabled enterprise transformation. "Insufficient analytical detail" referred to studies that mentioned digital platforms but did not provide enough information on transformation mechanisms, platform governance, stakeholder value, methods, or outcomes to support coding. These criteria were aligned with the eligibility criteria and were operationalized during full-text assessment.

2.2. Research process

2.2.1. Construction of the literature dataset

We searched Web of Science Core Collection and Scopus to build a cross-disciplinary dataset on digital platforms in enterprise contexts. The search combined platform-related terms, including "digital platform*", "digital ecosystem*", "digital marketplace*", "digital service platform*", "e-commerce platform*", "platform-based model*", "multi-sided platform*", and "platform as a service", with enterprise-related terms, including "enterprise", "firm", and "company". The search covered the period from database inception to 18 June 2025. Web of Science Core Collection and Scopus were selected because they provide broad cross-

disciplinary coverage and citation metadata suitable for bibliometric mapping. The complete search strings for both databases, including search fields, document type limits, language limits, and retrieval dates, are provided in Supplementary Material B.

The initial search returned 3,005 records. After deduplication, 2,467 unique records remained. Title and abstract screening and full-text eligibility assessment were conducted by two reviewers independently. Disagreements were resolved through discussion, with unresolved cases adjudicated by a third reviewer. Screening decisions were recorded in a shared screening log. At the title and abstract screening stage, 1,962 records were excluded because they did not address digital platforms in enterprise transformation contexts. A further 53 records were removed because the full text could not be retrieved. The remaining 452 articles were assessed in full text.

Studies were included when they met four criteria: (1) they examined digital platforms, digital platform ecosystems, or closely related platform-based models in enterprise contexts; (2) they addressed enterprise digital transformation or platform-enabled transformation mechanisms; (3) they provided empirical, modeling-based, case-based, or analytical evidence relevant to platform-enabled transformation; and (4) they were peer-reviewed English-language journal articles or book chapters. Studies were excluded when digital platforms were only marginally mentioned, enterprise transformation was outside the main focus, the publication was a review article, editorial, commentary, or non-scholarly source, or the full text was unavailable. The dataset was limited to peer-reviewed journal articles and book chapters to ensure traceable academic evidence and sufficient bibliographic metadata for bibliometric mapping. Reviews, editorials, commentaries, grey literature, and non-scholarly sources were outside the dataset.

In the final full-text assessment, 286 articles were excluded because DP-EDT was outside the main focus or because the publication was a review article. Fourteen articles were excluded because the full texts did not provide sufficient analytical detail to support extraction and coding. The full-text exclusion reasons were summarized by category in the screening record and PRISMA flow diagram. The final dataset comprised 152 peer-reviewed English-language journal articles and book chapters. These studies were used for both bibliometric mapping and qualitative content analysis. All included studies contributed to the bibliometric synthesis, and the qualitative synthesis coded them by platform type, research method, conceptual focus, stakeholder group, and value-related outcome.

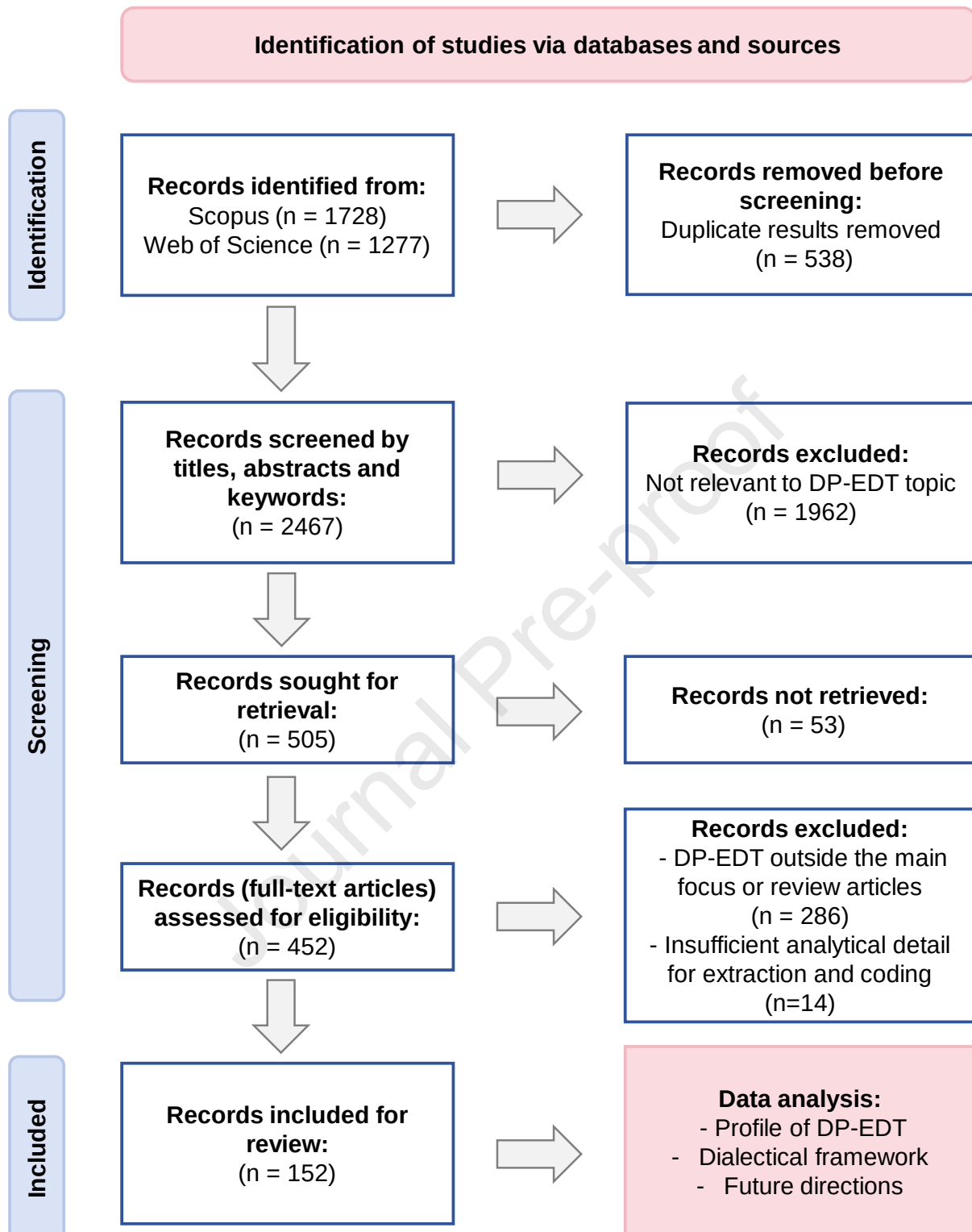


Fig. 1. PRISMA flow diagram

2.2.2. Analysis of the literature dataset

The literature dataset was prepared for bibliometric mapping and qualitative coding before analysis. Duplicate records were removed, and bibliographic fields were checked for consistency. Author keywords and Keywords Plus were standardized by merging spelling

variants, singular and plural forms, and common abbreviations. Missing bibliographic fields were retained as missing values and were not imputed. For qualitative coding, extracted variables were recoded into higher-level categories aligned with the concept, method, and value dimensions.

Bibliometric analysis was conducted using VOSviewer 1.6.20. We mapped journal sources, annual publication trends, regional distribution, and keyword co-occurrence networks to identify the structural patterns of DP-EDT research. Keyword networks were generated from author keywords and Keywords Plus, with a minimum occurrence threshold of five. The full-counting method was used, and clustering followed the modularity-based algorithm embedded in VOSviewer. These settings were selected to provide a stable overview of dominant themes and their relationships across the dataset.

Structured content analysis was then used to interpret the conceptual foundations, methodological orientations, and stakeholder value configurations of the included studies. Initial coding categories were derived from the research questions, bibliometric clusters, and the three analytical dimensions. Iterative reading was used to identify emergent subthemes. Through axial coding, we identified recurring patterns of tension and coordination across studies, including openness and control, efficiency and resilience, data sharing and privacy, and value creation and value distribution.

Results are presented through journal source tables, annual publication trends, regional distribution tables, keyword co-occurrence maps, method summary tables, platform-type classifications, stakeholder mapping, and the final dialectical systems framework. Tables were ordered by frequency or analytical category. Figures were organized to show structural patterns, conceptual relationships, and cross-dimensional coordination mechanisms.

No meta-analysis, quantitative effect synthesis, reporting bias assessment, or certainty-of-evidence grading was conducted. The included studies varied substantially in research design, sector, data source, platform type, and outcome definition. Statistical heterogeneity analysis, subgroup analysis, and meta-regression were therefore not applicable. Conceptual heterogeneity was examined qualitatively by comparing differences across platform type, sector, organizational context, stakeholder configuration, research method, and transformation outcome.

The final dialectical systems framework was developed by integrating the bibliometric and qualitative findings. The concept dimension captures the main components of DP-EDT, including drivers, platform types, transformation pathways, governance mechanisms, and effects. The method dimension captures how DP-EDT has been studied and evaluated. The value dimension captures stakeholder claims, conflicts, and coordination mechanisms. Together, these dimensions provide the analytical basis for explaining platform-enabled enterprise transformation as a process shaped by cross-level interaction and recurring tension.

3. Profile of DP-EDT research

3.1. Sources of publication

Table 1 presents the main publication sources contributing to DP-EDT research. The included studies were distributed across journals in knowledge economy, international

marketing, production economics, engineering management, sustainability, and technology management. This distribution shows that DP-EDT has developed as a cross-disciplinary research area, with contributions from technological, organizational, managerial, and economic perspectives.

The publication pattern is dispersed. No single journal dominates the field, and the most frequent source accounts for only a small share of the dataset. This dispersion indicates that DP-EDT research has not yet formed a consolidated disciplinary home. It also helps explain why existing studies often approach platform-enabled transformation through separate lenses, such as digital capability, business model innovation, supply chain coordination, internationalization, and ecosystem governance.

A list of included studies is provided in Supplementary Material C. No study-level risk of bias or methodological quality table is presented because no formal study-level risk of bias assessment was conducted. The limitations of the included evidence are summarized at the review level in the Discussion.

Table 1. Sources of publications

Journal	Number	Percentage
<i>Sustainability (Switzerland)</i>	7	4.61%
<i>Technological Forecasting and Social Change</i>	4	2.63%
<i>Journal of the Knowledge Economy</i>	3	1.97%
<i>International Marketing Review</i>	3	1.97%
<i>International Journal of Production Economics</i>	3	1.97%
<i>IEEE Transactions on Engineering Management</i>	3	1.97%
<i>Industrial Marketing Management</i>	2	1.32%
<i>Journal of Agribusiness in Developing and Emerging Economies</i>	2	1.32%
<i>Journal of World Business</i>	2	1.32%
<i>Studies in Science of Science</i>	2	1.32%

3.2. Publication period distribution

Fig. 2 illustrates the publication trends of DP-EDT research over the past decade. From 2012 to 2017, the field remained in an early exploratory stage, characterized by low and relatively stable publication activity. Between 2018 and 2021, research output began to grow steadily, reflecting the gradual expansion of digital platform applications and the increasing

importance of enterprise digital transformation. During this period, scholarly interest increased, although growth remained moderate.

A sharp increase occurred in 2022, likely driven by the acceleration of digital practices in response to the COVID-19 pandemic. This surge highlights the catalytic role of global disruptions in prompting practical adoption and academic inquiry. From 2023 to 2025, research activity has consistently remained strong. Preliminary data from the first half of 2025 indicate continued momentum, suggesting that the DP-EDT remains a vibrant and evolving research focus.

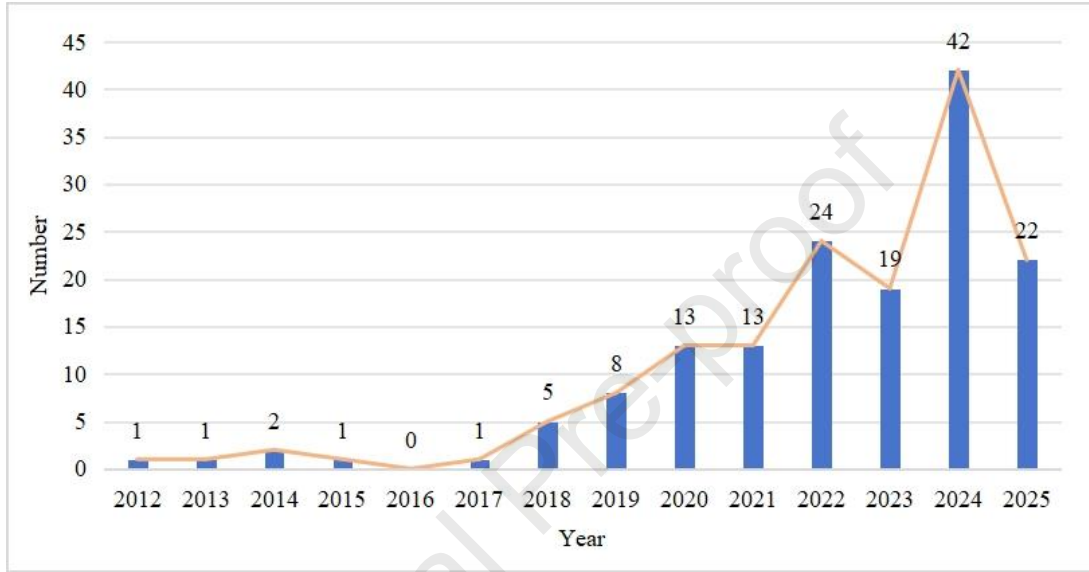


Fig. 2. Annual number of publications

3.3. Distribution of publications

Table 2 presents the regional distribution of publications on DP-EDT. The data reveal clear spatial patterns in research output. China and the United States lead this field, contributing nearly 40% of the total publications. Their dominant positions reflect the maturity of their digital economies and the depth of their enterprise transformation practices.

In Europe, research is primarily concentrated in countries such as Germany, the Nordic nations, and Italy, highlighting the region's longstanding strengths in industrial digitalization. Although Latin America, Africa, and Oceania contribute a smaller share, emerging studies from these regions suggest the development of distinct research trajectories shaped by local contexts and digital adoption patterns.

Table 2. Distribution of publications by region

Region	Country	Number	Percentage
Asia	China	35	23.00%
	India	10	6.60%
	Indonesia	5	3.30%
	Malaysia	4	2.60%
	Vietnam	3	2.00%

	Japan	2	1.30%
	Germany	12	7.90%
	Italy	8	5.30%
Europe	United Kingdom	6	3.90%
	France	5	3.30%
	Nordic countries	9	5.90%
	The United States	25	16.40%
Americas	Brazil	5	3.30%
	Ecuador	3	2.00%
Africa	South Africa	3	2.00%
	Nigeria	2	1.30%
Oceania	Australia	4	2.60%
Others		11	7.30%

3.4. Keyword co-occurrence network

Fig. 3 presents the keyword co-occurrence network of DP-EDT research. The network is centered on two high-frequency terms: “digital transformation” and “digital platforms”. Their central position indicates that the field is organized around the relationship between platform-based digital infrastructure and enterprise-level transformation.

The first cluster concerns technological infrastructure and organizational capability. Keywords such as “cloud computing” and “artificial intelligence” show that existing studies often treat digital platforms as technical infrastructures that support data integration, process automation, and capability development. This cluster reflects the technology-enabled foundation of DP-EDT.

The second cluster concerns sectoral application and industry-specific transformation. Keywords such as “automotive industry” and “tourism” indicate that platform-enabled transformation has been examined in different industrial settings. These studies emphasize how platform functions are shaped by sectoral conditions, market structures, and operational requirements.

The third cluster concerns innovation, sustainability, and long-term strategic change. Keywords such as “open innovation” and “sustainable development” show that digital platforms are increasingly linked with business model renewal, ecosystem collaboration, and broader transformation outcomes. This cluster extends the field from technology adoption and operational improvement to strategic and societal implications.

Overall, the keyword network shows that DP-EDT research has developed along three connected lines: technological enablement, sectoral application, and strategic transformation. This structure supports the subsequent qualitative synthesis by showing why DP-EDT needs to be examined across concept, method, and value dimensions.

deliver value (Jeza and Lekhanya, 2022; Xia and Gao, 2018). Third, institutional and policy support create enabling conditions for transformation. National strategies for data sharing, digital infrastructure, and industry upgrading reduce adoption barriers and encourage firms to invest in platform-enabled transformations (Li and Mei, 2024).

From an internal perspective, DP-EDT is shaped by three main organizational motivations. The first is efficiency improvement. Firms adopt digital platforms to streamline operations, reduce redundancy, and improve coordination across business functions, especially when resource integration and process visibility are critical (Hermes et al., 2020; Simonsson et al., 2020). The second is innovation and capability-building. Digital platforms can function as innovation infrastructures that connect firms with external knowledge, technical resources, and collaborative partners, thereby strengthening learning, experimentation, and new service development (Romero and Mammadov, 2025; Sassanelli and Terzi, 2022). The third is competitiveness and strategic renewal. Firms increasingly use platforms not only to optimize existing operations but also to develop new value propositions, enter new markets, and reposition themselves within broader platform ecosystems (Schmidt et al., 2019). Taken together, these external and internal drivers suggest that DP-EDT is propelled not by a single factor but by the interaction between environmental change and firm-level strategic response.

4.1.2. Digital platform types of DP-EDT

With the rapid advancement of the digital economy, digital platforms play an increasingly important role in industrial upgrading, resource allocation, and business model transformation. Different platform types have emerged across enterprise contexts, each with distinct functional emphasis and governance implications. These differences reflect not only technological architectures but also strategic intent and operational logic.

4.1.2.1 Technology-assisted platforms

Technology-assisted platforms are centered on advanced digital technologies and primarily support operational upgrading. By integrating cloud computing, big data, artificial intelligence, and the Internet of Things, they can improve efficiency, innovation capacity, and responsiveness (Yu and Ma, 2025). Their main contributions lie in lowering the threshold for technology deployment and enabling business reconfiguration through digital infrastructure (Ruutu et al., 2017; Shen et al., 2023). In industrial settings, platforms such as COSMOPlat and Predix support more precise control and flexible production, whereas cloud-based platforms reduce the cost of enterprise informatization and provide scalable computing resources (Liu et al., 2023a; Nayak et al., 2025).

4.1.2.2 Information-matching platform

Information-matching platforms primarily connect supply and demand and support information sharing, resource matching, and value creation in B2B and B2C settings (Arrigo, 2022). E-commerce platforms such as Amazon and Alibaba reduce transaction friction by connecting dispersed actors and increasing market visibility (Da Rocha et al., 2024; Hu et al., 2024). Similar functions appear in logistics and agricultural data-sharing platforms, where coordination efficiency depends on network effects and data aggregation (Budde et al., 2024; Da Rocha et al., 2024).

4.1.2.3 Transaction settlement platform

Transaction settlement platforms extend beyond matching by integrating payments, logistics, and transaction completion into a single-platform environment. Their main functions are to reduce transaction costs and enable end-to-end value delivery. Ride-hailing, food delivery, and B2B market platforms are typical examples, as they combine matching functions with settlement and service execution (Deveney, 2018; Jean et al., 2020; Rawling and Munton, 2022).

4.1.2.4 Innovation collaboration platforms

Innovation collaboration platforms support joint problem-solving, knowledge sharing, and collaborative innovation across organizations. They are widely used in design, R&D, and supply chain collaboration. Their value lies in connecting distributed expertise and accelerating experimentation, co-development, and knowledge recombination (Belloc, 2025; Hasselwander, 2024; Liu et al., 2022).

4.1.2.5 Internal management and communication platforms

Internal management and communication platforms focus on intra-organizational digitalization. They support information flow, process automation, and cross-functional coordination through enterprise systems, such as ERP, CRM, and enterprise social platforms (Caputo et al., 2022; Cenamor et al., 2019; Zhang et al., 2025). Their main contribution lies in integrating internal business functions and improving organizational visibility and control.

The literature suggests that platform transformation does not follow a single model. Different platform types support different strategic priorities, and firms often combine multiple platform logics over time. Simultaneously, these platform types embody recurring dialectical tensions. Technology-assisted and internal management platforms tend to prioritize control, standardization, and internal integration, whereas innovation collaboration and information-matching platforms depend more heavily on openness, participation, and decentralized interaction. Transaction settlement platforms emphasize efficiency and closure but may reduce adaptability when conditions change.

These tensions are often easier to describe conceptually than to examine empirically. Many existing studies classify platform types by function but do not examine how firms move across types, how governance logic shifts over time, or how these trade-offs are operationalized in research design. As a result, tensions between openness and control or between efficiency and adaptability are often treated as background assumptions rather than measurable transformation processes.

These tensions also generate stakeholder-level conflicts and coordination needs. More open platform types create stronger demands for trust, participation rules, and dispute resolution among firms, users, and complementors, whereas more closed platform types redistribute authority to platform operators and raise concerns about dependence, exclusion, and limited flexibility. In this sense, platform type is not only a conceptual category but also a source of value conflict and governance coordination across the broader ecosystem.

4.1.3. Path research on DP-EDT

This study identified five core pathways for DP-EDT, as shown in Fig. 4. Rather than representing five independent routes, these pathways form an evolving transformation logic through which digital platforms support enterprises at different stages of digital transformation.

Together, they demonstrate how platforms enable technological upgrading, business model reconfiguration, market expansion, ecosystem collaboration, and resilience enhancement.

Existing studies suggest that these pathways should be understood not only as thematic categories but also as processual patterns. In the early stages of transformation, firms primarily rely on digital platforms to establish digital connectivity, optimize internal processes, and improve operational visibility. As the transformation deepens, platforms play a stronger role in business model reconfiguration, market expansion, and external resource integration. In the more mature stages, the focus shifts toward ecosystem orchestration, governance balancing, and resilience building. These priorities also vary across organizational contexts: start-ups often emphasize experimentation and market access; incumbent firms focus more on integration and reconfiguration; and traditional industries face stronger tensions related to path dependence, coordination complexity, and governance adaptation. This processual perspective explains why the same platform may support different transformation priorities over time.

With the rapid development of digital technology, digital platforms have become core drivers of enterprise digital transformation (Masuda et al., 2021). Existing studies show that platforms provide structured transformation logic through technological enablement, process optimization, resource integration, and collaborative innovation, thereby improving competitiveness, innovation capacity, and long-term adaptability in complex environments (Schmidt et al., 2019; Lerch and Heimberger, 2022).

4.1.3.1 Technology assistance and process optimization path

Digital platforms integrate advanced technologies, such as cloud computing, big data, artificial intelligence, and the Internet of Things, to provide enterprises with powerful technological toolkits, directly enhancing internal operational efficiency (Evans et al., 2022; Okano et al., 2021). They support the automation and intelligence of core processes such as production, supply chain, procurement, and finance, thereby reducing manual intervention, error rates, and operational costs (Lerch and Heimberger, 2022). By integrating internal and external data, platforms enable more precise and timely decision-making (Masuda et al., 2021; Zhao and Zha, 2024). In addition, they improve collaborative efficiency by connecting suppliers, customers, and partners and by increasing the visibility of information, logistics, and capital flows (Heinbach et al., 2022; Marzi et al., 2023).

4.1.3.2 Innovation-driven and business model reconfiguration path

Platforms are not only efficient tools but also catalysts for innovation and business model change (Lerch and Heimberger, 2022; Pauli et al., 2025). Their openness, modularity, and connectivity facilitate knowledge sharing, technical exchange, and collaborative R&D with external partners, thereby accelerating the development of new products, services, and technologies (Pauli et al., 2025; Xing et al., 2024). Platforms also support transitions from product-centered models to more service-oriented and platform-based value creation logic (Nwaiwu, 2021; Zhou et al., 2023). Through external knowledge search and flexible resource recombination, firms can strengthen their innovation performance and respond more effectively to emerging opportunities (Zhao and Zha, 2024).

4.1.3.3 Market expansion and customer value-deepening paths

Platforms have restructured the relationship between enterprises and markets, thus

becoming a critical channel for competitive positioning (Jeza and Lekhanya, 2022). In global market access and digital marketing, B2B and B2C platforms reduce entry barriers and transaction costs while supporting targeted positioning, personalized recommendations, and more effective promotion strategies (Simonsson et al., 2020). Platforms also enable deeper customer interaction by facilitating real-time feedback and demand sensing, which helps firms adjust their products, services, and marketing strategies more quickly (Nayak et al., 2025; Pauli et al., 2025). Additionally, they support customized production and customer participation in value co-creation through more flexible supply chains and partner networks (Marzi et al., 2023; Zghurska et al., 2022).

4.1.3.4 Ecosystem construction and collaborative symbiotic paths

The central value of digital platforms lies in their capacity to build and coordinate ecosystems (Li and Mei, 2024). Platforms connect enterprises, suppliers, customers, complementors, and even competitors into broader value co-creation networks, thereby facilitating the flow of knowledge, technology, data, and resources across organizational boundaries (Xing et al., 2024). Through standardized interfaces and governance rules, they reduce collaboration costs and support cross-enterprise and cross-industry cooperation (Kim, 2024). They are also particularly important in helping small and medium-sized enterprises (SMEs) gain access to external resources and participate in ecosystems that would otherwise be dominated by larger firms (Gusenbauer et al., 2025; Xia and Gao, 2018).

4.1.3.5 Resilience enhancement and sustainable development paths

Platforms can strengthen environmental sensing and help firms monitor market dynamics, policy changes, and supply chain risks in real time. Combined with data analytics, this supports early warnings, contingency planning, and faster responses to uncertainty (Le et al., 2024). By improving information processing, resource allocation, and network flexibility, platforms can also help firms recover more quickly from disruptions (Liu et al., 2023c). In addition, they can support greener procurement, more efficient resource use, and sustainable business practices, thereby improving environmental and social performance while responding to regulatory and market expectations (Nwaiwu, 2021).

Taken together, these five pathways are better understood as evolving sequences of transformation priorities than as isolated categories. Early pathways are often associated with digital entry and process upgrading; intermediate pathways with business model renewal and market expansion; and later pathways with ecosystem governance and resilience. This sequencing is not linear or universal but highlights how platform-enabled transformation develops through shifting combinations of technological, organizational, and relational changes. Digital platforms do not simply offer multiple options; they structure how firms progress through different phases of transformation under different governance conditions.

In summary, digital platforms support digital transformation through multiple interconnected pathways, including technological enablement, process optimization, resource integration, business model innovation, knowledge coordination, and risk mitigation. These pathways improve operational efficiency and market competitiveness while also shaping long-term organizational adaptation. However, each pathway embodies a core dialectical tradeoff: efficiency versus resilience, openness versus control, short-term performance versus long-term

sustainability, and autonomy versus ecosystem dependence. Therefore, enterprises do not simply choose a single pathway. They move across pathways over time and continuously rebalance competing demands as the transformation progresses.

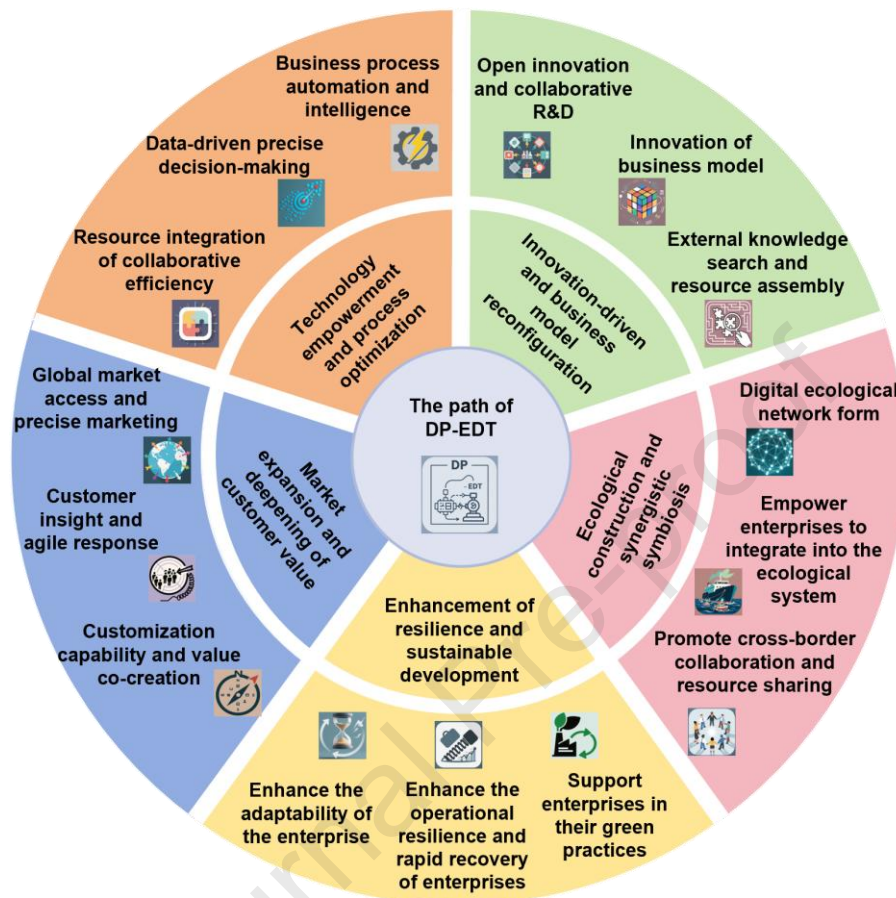


Fig. 4. Pathways through which digital platforms shape enterprise digital transformation.

4.1.4. Operation and management of DP-EDT

In the context of digital transformation, digital platforms serve as a core infrastructure for business reconfiguration, resource integration, and value creation. Their operation and management directly influence whether the platform-enabled transformation can be sustained over time.

4.1.4.1 Stakeholder collaboration and ecosystem governance.

The operation of digital platforms depends on the management of stakeholder relationships and the construction of collaborative ecosystems (Gusenbauer et al., 2025). Transparent governance structures, participation rules, evaluation systems, and dispute resolution mechanisms are required to support interaction and value creation among diverse actors (Kim, 2025).

4.1.4.2 Platform architecture design and user experience optimization

The platform architecture is a basic condition for operation and scalability. Flexible technical, business, and data architectures allow the platform to adapt to changing enterprise

needs and market conditions, whereas user-centered design supports adoption, satisfaction, and continued participation (Isensee et al., 2025; Lerch and Heimberger, 2022; Romero and Mammadov, 2025).

4.1.4.3 Data governance and information security assurance.

Data governance is central to platform operations because data constitute a key strategic asset. Effective operations depend on ensuring data quality, availability, security, and accountability through access control, encryption, and clearly defined governance responsibilities (Civelek et al., 2023; Heinbach et al., 2022; Zhao and Zha, 2024).

4.1.4.4 Partnership and ecosystem construction

Partnership management is another core element of platform operation. Stable partnerships allow platforms to extend their functional scope, integrate complementary resources, and enhance service capability across sectors, such as agriculture, manufacturing, and energy (Gusenbauer et al., 2025; Vodianka and Yurii, 2020).

4.1.4.5 Operational processes and performance evaluation mechanisms

Operational routines and evaluation systems are required to monitor platform performance, identify problems, and support continuous adjustment. Service standards, charging mechanisms, user feedback systems, and performance indicators contribute to the long-term viability of platform operations (Da Rocha et al., 2024; Lerch and Heimberger, 2022).

4.1.4.6 Organizational capacity building and digital literacy improvement.

Platform operations also depend on internal organizational capacity. Firms need digital skills, platform management capabilities, and strategic alignment between platform use and long-term organizational goals (Civelek et al., 2023; Evans et al., 2022).

Taken together, platform operations and management involve a coordinated set of governance, technical, organizational, and evaluative mechanisms. The central challenge is not simply how to optimize each component but also how to balance competing operational logic. Openness can attract participation and stimulate value creation. However, excessive openness may weaken platform order and security. Standardization improves consistency and efficiency but may reduce flexibility and responsiveness to diverse user needs. Data sharing supports collaboration and innovation but also raises concerns about privacy, security, and accountability. Cost and efficiency pressures can strengthen short-term performance while weakening long-term ecosystem capabilities, user trust, and organizational learning.

These governance trade-offs are frequently acknowledged in the literature, but are still only partially captured by dominant empirical designs. Existing studies often discuss them as conceptual or managerial challenges, while offering limited measurement of how such trade-offs evolve across platform stages, vary across organizational settings, or reshape governance outcomes over time. This makes operation and management a conceptually rich but methodologically underdeveloped area in DP-EDT research.

At the stakeholder level, these tensions redistribute control, privacy, participation, and risk across the groups. Greater openness may benefit complementors and users, but reduce platform operators' control; tighter governance may stabilize operations, but increase dependence on participating firms; stronger data extraction may improve coordination, but intensify privacy concerns for customers and compliance concerns for regulators. Platform operation is,

therefore, not only a managerial issue but also a value-allocation process that generates recurring coordination needs across the ecosystem.

4.1.5. *Effect of DP-EDT*

Digital platforms have become central to enterprise digital transformation and are widely associated with multiple forms of organizational and ecosystem-level impacts. Existing studies show that platform-enabled transformation influences not only internal performance but also broader patterns of innovation, resource allocation, market access, and social value creation.

4.1.5.1 Enhancing enterprises' operational efficiency

Research confirms the positive impact of digital platform capabilities on enterprise performance. By effectively utilizing digital platforms, enterprises can achieve significant improvements in operational efficiency, specifically manifested in cost reduction and improvement of product quality and service levels (Le et al., 2024). For example, enterprises can achieve cross-departmental collaboration, information sharing, and process automation through platforms, effectively breaking information silos and improving overall operational efficiency (Okano et al., 2021).

4.1.5.2 Improving the market competitiveness of enterprises

Digital platforms support enterprises in expanding their market opportunities and building strategic advantages. Through digital platforms, enterprises can overcome geographical limitations and collaborate with global suppliers to better meet the market's demand for customized products (Xing et al., 2024). For instance, cooperation with green suppliers can help enterprises respond to consumer demands for sustainable products, creating a competitive differentiation advantage. The application of B2B platforms enables enterprises to conduct transactions with global partners conveniently and significantly expand their market share (Marzi et al., 2023). This improvement in operational efficiency and expansion of market boundaries jointly enhance the market competitiveness of enterprises, becoming the core value manifestation of DT.

4.1.5.3 Enterprise innovation capabilities

Digital platforms provide enterprises with new tools, environments, and ecological support, thus becoming the core engine driving the enhancement of enterprise innovation capabilities. On the one hand, platforms provide a technical foundation for internal innovation, helping enterprises develop new products and services and explore new business models. On the other hand, they build an open collaborative innovation network, enabling enterprises to efficiently collaborate with external partners and accelerate the development and application of new technologies and products (Le et al., 2024).

4.1.5.4 Promoting enterprise business model transformation

At the business model level, digital platforms drive a fundamental transformation in the logic of enterprises' value creation. Traditional enterprises focus on maximizing consumer value, whereas platform enterprises focus on maximizing the ecosystem value. This transformation reconfigures the value chain of enterprises (Drewel et al., 2021). In the energy sector, bilateral market connection, project financing, and user-constructed digital platforms solve the problem of high initial investment for users and create stable returns for investors, forming a new revenue and value creation model (Nwaiwu, 2021). In the service sector, digital

platforms enable SMEs to undertake outsourcing business, breaking organizational boundaries to access global resources and expertise and achieve global task allocation (Gusenbauer et al., 2025).

4.1.5.5 Optimizing enterprise resource allocation

The role of digital platforms in optimizing resource allocation is mainly to improve resource utilization efficiency and achieve dynamic adjustment. Through the real-time monitoring function of the platform, enterprises can accurately grasp resource usage and make flexible adjustments according to market demand and business development, thereby achieving optimal resource allocation (Le et al., 2024). This dynamic configuration ability is particularly prominent in industrial chain collaborations. Digital platforms promote the collaborative development of large, small, and medium-sized enterprises, enhance the stability and efficiency of industrial and supply chains, and provide support for enterprises to reduce costs, improve production efficiency, and improve product quality by releasing data value (Li and Mei, 2024).

4.1.5.6 Enhancing an enterprise's risk resistance capability

Digital platforms significantly enhance an enterprise's risk resistance capability and resilience in a complex and volatile market environment. Through data sharing and collaborative cooperation, enterprises can respond more effectively to market fluctuations, raw material shortages, and other risks, ensuring operational stability (Lerch and Heimberger, 2022). In crisis response, the platform provides enterprises with stronger adaptability and planning capabilities, enabling them to recover and achieve long-term development quickly, thus reducing the negative impact of crises on profitability (Liu et al., 2023c). Additionally, digital platforms in cross-border transactions reduce information asymmetry and middlemen through data sharing, lowering export, marketing, and operational costs, further enhancing enterprises' risk resistance capability in the international market (Da Rocha et al., 2024).

4.1.5.7 Generating diverse economic impacts

Digital platforms that facilitate enterprises' DT not only impact the enterprises themselves but also release diverse values at the macroeconomic and social levels. DT reduces service costs and creates new business opportunities, positively affecting overall economic growth (Loffredo and Tufo, 2018). For example, the DT of the Ukrainian agricultural sector is expected to increase the country's GDP (Vodianka and Yurii, 2020). For SMEs, DT helps them gain opportunities to compete with large enterprises, promoting the enhancement of market vitality (Jeza and Lekhanya, 2022). Simultaneously, industrial collaboration driven by digital platforms forms new industrial clusters, promoting the upgrading of industrial structures and economic sustainability (Xing et al., 2024).

4.1.5.8 Enhancing an enterprise's social value

Digital platforms may also improve environmental performance and social outcomes by supporting greener procurement, better service access, and more efficient resource use, although the strength and distribution of these effects vary across sectors and governance contexts (Hermes et al., 2020; Marzi et al., 2023).

These effects show that DP-EDT does not generate a single outcome profile. Instead, platform-enabled transformation produces a bundle of economic, organizational, and social effects that often coexist with new trade-offs. Efficiency gains can improve short-term

performance but increase path dependence and reduce adaptability. Innovation may expand through platform openness, yet intensify tensions over control and resource allocation. Market growth and economic upgrading may be accompanied by new concerns about fairness, privacy, exclusion, and environmental burdens. In this sense, the effects of DP-EDT are not uniformly positive or linear but are shaped by how the competing outcomes are balanced over time.

It is also difficult to evaluate these tensions using dominant research designs. Much of the existing literature measures effects through firm-level performance indicators, adoption outcomes, or self-reported capability gains while paying less attention to cross-level consequences, distributional effects, and temporal trade-offs. As a result, many outcome studies identify whether digital platforms create value but provide weaker evidence on how value is distributed, what costs accompany it, and under what governance conditions positive effects are sustained.

At the stakeholder level, this creates recurring conflicts over whose outcomes count and whose risks are absorbed. Enterprises may prioritize efficiency and growth, customers may be more concerned with service quality and privacy, regulators may focus on fairness and legitimacy, and smaller firms may experience both empowerment and dependence. Therefore, the effects of DP-EDT generate not only performance outcomes but also coordination problems concerning value distribution, accountability, and long-term sustainability.

These concepts are summarized in Fig. 5. Therefore, Fig. 5 should be understood not merely as a static aggregation of concepts but as a structured representation of how drivers, platform types, pathways, governance mechanisms, and effects are connected through tension and adjustment processes. Existing research shows that digital platforms support enterprise DT through multiple and interrelated effects at both the firm and system levels. However, these effects are best interpreted as contingent outcomes shaped by evolving governance arrangements, measurement choices, and stakeholder negotiations, rather than as universally positive consequences of platform adoption.

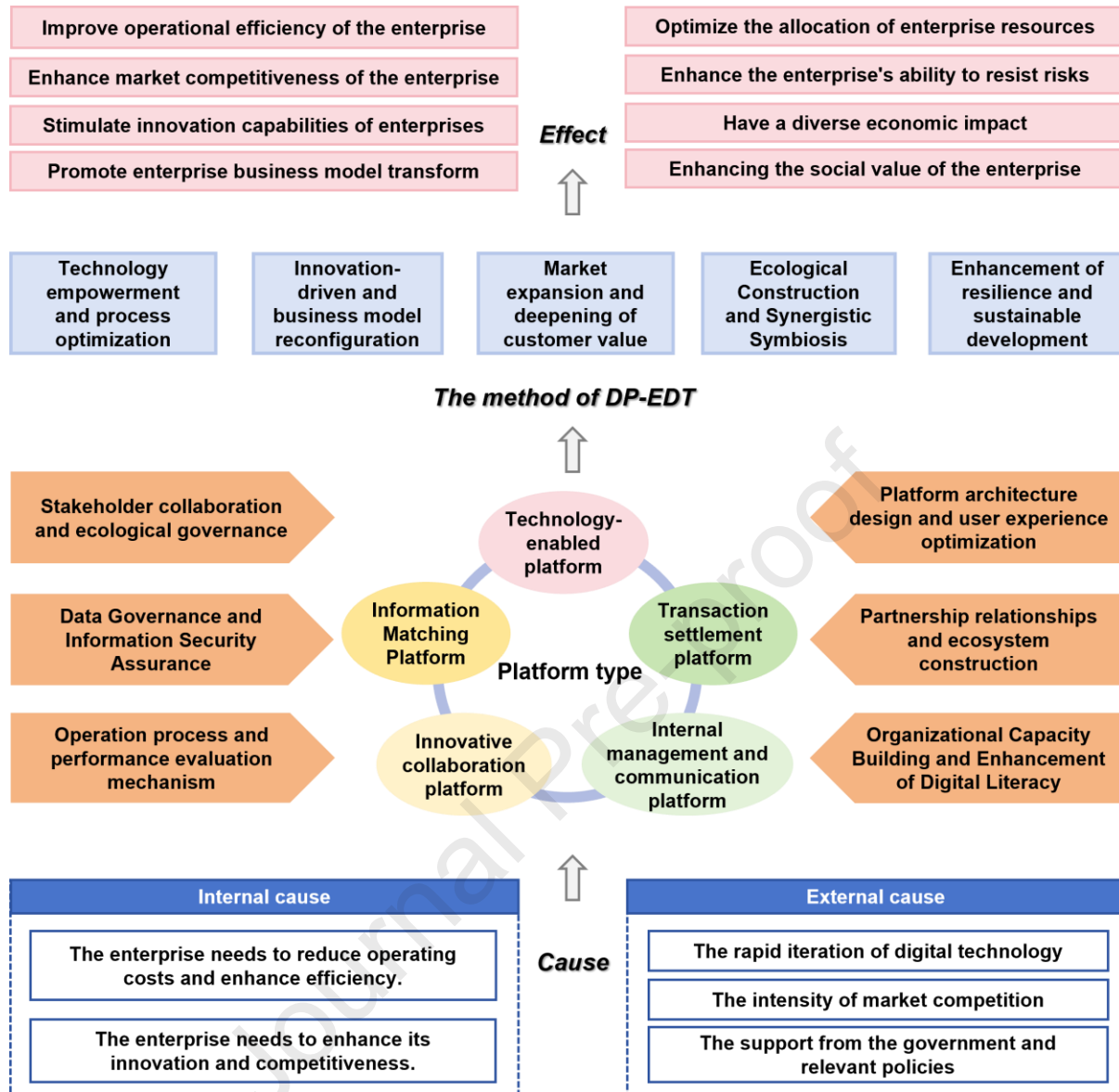


Fig. 5. Core concepts of digital platform-enabled enterprise digital transformation.

4.2. Methodological dimension

The methodological literature on DP-EDT is not defined by methodological diversity alone but by uneven patterns of application across research themes. Existing studies differ not only in the methods they use but also in the types of transformation processes they are able to capture. From a dialectical perspective, the methodological dimension is therefore important not merely as a technical choice but also as a condition that shapes what kinds of tensions, mechanisms, and outcomes become visible in the literature.

4.2.1. Quantitative research patterns in DP-EDT

Quantitative research on DP-EDT is primarily used to examine firm-level relationships among platform capabilities, strategic orientation, innovation, and performance outcomes. Rather than being evenly distributed across research themes, quantitative methods are concentrated in studies that test the direct, mediating, and moderating relationships among

organizational variables. Survey-based designs, especially those using structural equation modeling and regression techniques, are the dominant approach in this stream (Ahmed et al., 2022; Caputo et al., 2022; Cenamor et al., 2019; Jean et al., 2020). These studies typically focus on how digital platform capabilities, digital culture, supply chain innovation, and business model innovation influence enterprise performance and transformation.

The second quantitative stream adopts modeling and simulation methods to capture platform dynamics at the system level. Studies using system dynamics, evolutionary game theory, and mathematical modeling are less common, but they are particularly relevant for examining strategic interactions, governance evolution, and resource allocation in platform ecosystems (Huang and Tang, 2025; Ruutu et al., 2017). Compared with survey-based work, this stream is more attentive to interdependence and adaptation, but remains relatively limited in volume.

The third pattern is the use of panel data, descriptive statistics, and comparative quantitative techniques to identify temporal variations, heterogeneity, and basic empirical regularities. Panel data studies assess the long-term effects of platform use on enterprise outcomes, while descriptive statistics and group comparison methods are more often used to profile platform adoption, transformation challenges, and performance differences across firms (Huo et al., 2024; Ibarra-Fretell and Grijalva-Salazar, 2025; Xu and Sukpasjaroen, 2024). However, these approaches are still less common than cross-sectional survey designs.

Taken together, the quantitative literature shows a clear methodological pattern: firm-level capability–performance studies are dominated by survey, SEM, and regression approaches, whereas system-level dynamics and temporal evolution receive much less attention (as shown in Table 3). This imbalance helps explain why the field has generated relatively strong evidence of static organizational relationships but more limited evidence on cross-level tensions, evolving coordination mechanisms, and long-term transformation processes.

Table 3. Summary of quantitative research methods

Data collection methods	Applications in platform type	Applications in core concept
		The path of digital platforms facilitating DT of enterprises (Bhatti et al., 2025; Shetty and Panda, 2023; Xie et al., 2022)
		Operation and management of digital platforms (Xie et al., 2022)
Structural equation model (SEM)	Technology-enabled platform	The impact of digital platforms on enterprises (Caputo et al., 2022; Shetty and Panda, 2023)
		Analysis of factors contributing to the DT of enterprises through digital platforms (Nayak et al., 2025)
		The impact of digital platforms on enterprises (Ahmed et al., 2022; Suryani et al., 2021; Wang et al., 2014; Zghurska et al., 2022)

Regression analysis	Information matching platform	The operation and management of digital platforms (Zghurska et al., 2022)
		Analysis of factors contributing to DT of enterprises through digital platforms (Gong et al., 2024)
		The path of digital platforms facilitating the DT of enterprises (Bhatti et al., 2025)
		The operation and management of digital platforms (Liu et al., 2023b)
		The impact of digital platforms on enterprises (Liu et al., 2023c, 2024)
	Trading settlement platform	The impact of digital platforms on enterprises (Liu et al., 2023b)
		Analysis of the factors contributing to the DT of enterprises through digital platforms (Gong et al., 2024)
		The path of digital platforms facilitating the DT of enterprises (Bhatti et al., 2025; Shetty and Panda, 2023; Xia and Gao, 2018; Xie et al., 2022)
		The operation and management of digital platforms (Caputo et al., 2022; Liu et al., 2023c, 2024)
		The impact of digital platforms on enterprises (Ahmed et al., 2022; Caputo et al., 2022; Liu et al., 2023c, 2024; Shetty and Panda, 2023)
Regression analysis	Innovative collaboration platform	The path of digital platforms facilitating the DT of enterprises (Jean et al., 2020)
	Internal management and communication platform	Analysis of the factors contributing to the DT of enterprises through digital platforms (Gong et al., 2024)
		The path of digital platforms facilitating the DT of enterprises (Cenamor et al., 2019; Zhang et al., 2025)
		The path of digital platforms facilitating the DT of enterprises (Hasselwander, 2024; Liu et al., 2023; Romero and Mammadov, 2025; Tse et al., 2023; Zhou, 2024)
		The impact of digital platforms on enterprises (Jiang et al., 2023; Wang et al., 2023; Zhao and Zha, 2024)
Regression analysis	Technology-enabled platform	The operation and management of digital platforms (Romero and Mammadov, 2025; Zhou, 2024)
		The path of digital platforms facilitating the DT of enterprises (Hasselwander, 2024; Singh and Kapoor,

	Trading settlement platform	2024) The path of digital platforms facilitating the DT of enterprises (Gong et al., 2024; Hasselwander, 2024; Singh and Kapoor, 2024) The impact of digital platforms on enterprises (Da Rocha et al., 2024)
	Internal management and communication platform	The path of digital platforms facilitating the DT of enterprises (Le et al., 2024)
One-way analysis of variance	Technology-enabled platform	The path of digital platforms facilitating the DT of enterprises (Huo et al., 2024) The operation and management of digital platforms (Huo et al., 2024)
	Information matching platform	The path of digital platforms facilitating the DT of enterprises (Huo et al., 2024)
System dynamics	Information matching platform	The path of digital platforms facilitating the DT of enterprises (Ruutu et al., 2017)
	Trading settlement platform	The path of digital platforms facilitating the DT of enterprises (Ruutu et al., 2017)
Panel data analysis	Technology-enabled platform	The impact of digital platforms on enterprises (Xu and Sukpasjaroen, 2024; Yu and Ma, 2025)
Descriptive statistics	Innovative collaboration platform	The path of digital platforms facilitating the DT of enterprises (Shen et al., 2023; Xie et al., 2024) The operation and management of digital platforms (Shen et al., 2023) The impact of digital platforms on enterprises (Xie et al., 2024)
		The path of digital platforms facilitating the DT of enterprises (Ibarra-Fretell and Grijalva-Salazar, 2025; Jeza and Lekhanya, 2022; Ji et al., 2025; Pauli et al., 2025; Shen et al., 2023; Vodianka and Yurii, 2020; Xing et al., 2024)
		The impact of digital platforms on enterprises (Civelek et al., 2023; Xie et al., 2024)
Evolutionary game theory	Innovative collaboration platform	The path of digital platforms facilitating the DT of enterprises (Huang and Tang, 2025)
	Technology-enabled platform	The impact of digital platforms on enterprises (Li and Mei, 2024) The operation and management of digital platforms (Li and Mei, 2024)

Mathematical modeling	Technology-enabled platform	The impact of digital platforms on enterprises (Chen et al., 2015; Hermes et al., 2020; Schreieck et al., 2024)
	Information matching platform	The operation and management of digital platforms (Kim, 2025)
		The impact of digital platforms on enterprises (Chen et al., 2015; Schreieck et al., 2024)

Note: DT: digital transformation.

4.2.2. Qualitative research patterns in DP-EDT

Qualitative research in DP-EDT mainly addresses ecosystem formation, governance design, platform evolution, and innovation processes (Table 4). Case studies and semi-structured interviews are the dominant approaches in this stream. They are commonly used to trace organizational change, stakeholder interaction, and context-specific transformation mechanisms (Ben Arfi and Hikkerova, 2021; Bustamante, 2023; Ruggieri et al., 2018; Ruhland et al., 2021). These methods are particularly useful for examining how platform strategies unfold over time, how actors interpret transformation pressures, and how governance arrangements are negotiated in specific settings.

Other qualitative approaches, including content analysis, expert interviews, Delphi studies, comparative analysis, and pattern-based synthesis, appear less frequently and mainly support studies on platform governance, strategic design, and policy orientation (Brunetti et al., 2020; Drewel et al., 2021; Gusenbauer et al., 2025; Sauvagerd et al., 2024). These approaches strengthen the field by making process dynamics, stakeholder interests, and contextual conditions more visible. They also help explain how platform-related tensions emerge, persist, and become managed through formal rules, trust arrangements, data governance, and incentive alignment.

Within the DP-EDT literature defined by our review protocol, value-related analysis often relies on static constructs such as satisfaction, trust, and performance. Fairness, privacy, control, legitimacy, and accountability receive more limited integration in studies that jointly examine platform governance, enterprise transformation, and stakeholder value distribution. Adjacent literatures on platform ethics, financial inclusion, digital responsibility, information privacy, and data governance have examined these concerns more directly. The remaining gap in DP-EDT research lies in connecting these normative and governance concerns with enterprise transformation processes and platform-based value allocation.

Table 4. Review of qualitative research methods

Qualitative research methods	Applications in platform types	Applications in core concepts
Case study method	Technology-enabled platform	Analysis of the factors contributing to the DT of enterprises through digital platforms (Jia et al., 2024; Roberts and Kim, 2024) The path of digital platforms facilitating the DT of

		enterprises (Ben Arfi and Hikkerova, 2021; Budde et al., 2024; Bustamante, 2023; Da Rocha et al., 2024; Kim, 2024; Pauli et al., 2025; Roberts and Kim, 2024; Sassanelli and Terzi, 2022; Xiufan and Decheng, 2024)
		The operation and management of digital platforms (Rocha et al., 2023; Xiufan and Decheng, 2024)
		The impact of digital platforms on enterprises (Masuda et al., 2018, 2021)
		Analysis of the factors contributing to the DT of enterprises through digital platforms (Bustamante, 2023; Calabrese et al., 2021)
		The path of digital platforms facilitating the DT of enterprises (Da Rocha et al., 2024; Bustamante, 2023)
Information matching platform		The operation and management of digital platforms (Bustamante, 2023; Da Rocha et al., 2024; Mukhamedova et al., 2022; Rocha et al., 2023; Xiufan and Decheng, 2024)
		The impact of digital platforms on enterprises (Bustamante, 2023)
		The path of digital platforms facilitating the DT of enterprises (Okano et al., 2021)
Trading settlement platform		The operation and management of digital platforms (Bustamante, 2023; Da Rocha et al., 2024)
		The impact of digital platforms on enterprises (Kim, 2024; Mukhamedova et al., 2022)
		Analysis of the factors contributing to the DT of enterprises through digital platforms (Ben Arfi and Hikkerova, 2021)
Innovative collaboration platform		The path of digital platforms facilitating the DT of enterprises (Ben Arfi and Hikkerova, 2021; Budde et al., 2024; He and Sun, 2025)
		The operation and management of digital platforms (Mukhamedova et al., 2022)
		The impact of digital platforms on enterprises (Kovacic et al., 2020; Vodianka and Yurii, 2020)
Internal management and communication		The path of digital platforms facilitating the DT of enterprises (Haberly et al., 2019)
Semi-structured interview method	Technology-enabled platform	Analysis of the factors contributing to the DT of enterprises through digital platforms (Roberts and Kim, 2024; Ruggieri et al., 2018)
		The path of digital platforms facilitating the DT of

	enterprises (Ji et al., 2025; Ruggieri et al., 2018; Rocha et al., 2023)
	The operation and management of digital platforms (Drewel et al., 2021; Evans et al., 2022; Müller, 2019; Ruggieri et al., 2018)
	The impact of digital platforms on enterprises (Ruhland et al., 2021; Sauvagerd et al., 2024)
	Analysis of the factors contributing to the DT of enterprises through digital platforms (Fitz and Scheeg, 2025; Govik, 2024)
Information matching platform	The path of digital platforms facilitating the DT of enterprises (Li and Long, 2024; Liu et al., 2023b; Sauvagerd et al., 2024)
	The impact of digital platforms on enterprises (Drewel et al., 2021; Jeza and Lekhanya, 2022; Ruhland et al., 2021; Sauvagerd et al., 2024)
Trading settlement platform	The path of digital platforms facilitating the DT of enterprises (Arrigo, 2022; Fitz and Scheeg, 2025)
	The impact of digital platforms on enterprises (Drewel et al., 2021; Jeza and Lekhanya, 2022; Ruhland et al., 2021; Sauvagerd et al., 2024)
Innovative collaboration platform	The path of digital platforms facilitating the DT of enterprises (Schmidt et al., 2019)
	The operation and management of digital platforms (Schmidt et al., 2019)
	The impact of digital platforms on enterprises (Ruhland et al., 2021; Sauvagerd et al., 2024)
Internal management and communication	The operation and management of digital platforms (Drewel et al., 2021; Evans et al., 2022; Müller, 2019; Ruggieri et al., 2018; Schmidt et al., 2019)
Content analysis	The path of digital platforms facilitating the DT of enterprises (Müller, 2019; Shen et al., 2023; Wahyudiono et al., 2024)
	Technology-enabled platforms
	The operation and management of digital platforms (Babkin et al., 2022)
	The impact of digital platforms on enterprises (Albújar-Verona et al., 2022; Shen et al., 2023)
	The path of digital platforms facilitating the DT of enterprises (Babkin et al., 2022; Isensee et al., 2025)
	Information matching platform
	The operation and management of digital platforms (Babkin et al., 2022; Gusenbauer et al., 2025; Mukherjee and Wood, 2021)

Delphi method and expert interview	Trading settlement platform	Analysis of the factors contributing to the DT of enterprises through digital platforms (Marzi et al., 2023)
		The operation and management of digital platforms (Mukherjee and Wood, 2021)
		The impact of digital platforms on enterprises (Mukherjee and Wood, 2021)
	Innovative collaboration platform	The path of digital platforms facilitating the DT of enterprises (Calabrese et al., 2021)
		The operation and management of digital platforms (Gusenbauer et al., 2025)
		The impact of digital platforms on enterprises (Albújar-Verona et al., 2022; Mukherjee and Wood, 2021; Shen et al., 2023)
	Internal management and communication	The path of digital platforms facilitating the DT of enterprises (Hasselwander, 2024)
		Analysis of the factors contributing to the DT of enterprises through digital platforms (Gusenbauer et al., 2025)
	Technology- enabled platform	The path of digital platforms facilitating the DT of enterprises (Evans et al., 2022; Lerch and Heimberger, 2022)
		The operation and management of digital platforms (Evans et al., 2022; Lerch and Heimberger, 2022)
		The impact of digital platforms on enterprises (Brunetti et al., 2020)
Pattern-based approach and comparative	Technology- enabled platform	The path of digital platforms facilitating the DT of enterprises (Jean et al., 2020)
		The operation and management of digital platforms (Singh and Kapoor, 2024)
		The impact of digital platforms on enterprises (Hermes et al., 2020)
		The path of digital platforms facilitating the DT of enterprises (Gong et al., 2024)
		The path of digital platforms facilitating the DT of enterprises (Ben Arfi and Hikkerova, 2021; Jesus et al., 2024)
Pattern-based approach and comparative	Technology- enabled platform	The impact of digital platforms on enterprises (Dossymbek et al., 2025)
		Analysis of the factors contributing to the DT of enterprises through digital platforms (Vodianka and Yurii, 2020)

analysis method	The path of digital platforms facilitating the DT of enterprises (Huo et al., 2024; Wahyudiono et al., 2024)
	The operation and management of digital platforms (Vodianka and Yuri, 2020)
	The impact of digital platforms on enterprises (Ahmed et al., 2022)
Information matching platform	The path of digital platforms facilitating the DT of enterprises (Belloc, 2025)
	The operation and management of digital platforms (Deveney, 2018)
Innovative collaboration platform	The path of digital platforms facilitating the DT of enterprises (Liu et al., 2022)
	The operation and management of digital platforms (Liu et al., 2022; Schmidt et al., 2019)

Note: DT: digital transformation.

4.2.3. Methodological implications for dialectical analysis

These methodological patterns have direct implications for dialectical analysis. Static variable-based designs are useful for identifying relationships among organizational factors, but they are less effective at capturing evolving tensions, such as openness versus control, efficiency versus resilience, or data sharing versus privacy. By contrast, qualitative and process-oriented designs are more capable of revealing how such tensions are negotiated, how coordination mechanisms emerge, and why similar platform configurations generate different outcomes across contexts.

Therefore, from a dialectical systems perspective, methodological choice is not neutral. It shapes which dimensions of transformation are visible and which remain obscured. Quantitative methods tend to privilege measurable organizational relationships, whereas qualitative approaches are better suited to uncovering contextualized conflict–coordination dynamics. Future research could benefit from staged, longitudinal, and integrated designs that combine broad empirical coverage with process-level explanations. Such designs are more likely to capture the multi-level and evolving nature of DP-EDT and connect conceptual tensions with stakeholder value outcomes.

In this sense, the methodological dimension does not merely support empirical analysis but also mediates how conceptual structures and value conflicts are interpreted. Therefore, it provides a crucial bridge between the concept and value dimensions of dialectical frameworks.

4.3. Value dimension

The value dimension of DP-EDT concerns how different actors define, pursue, and negotiate value within a platform-enabled transformation. Instead of treating stakeholders as separate beneficiaries with independent interests, this review conceptualizes the value dimension as a field of competing claims, recurring conflicts, and coordination mechanisms. Across the literature, value creation in digital platform ecosystems is shaped by the interactions

among enterprises, customers, service providers, platform operators, regulators, researchers, and public actors. What matters is not only what each actor seeks but also how these claims come into tension and how they are coordinated over time.

4.3.1. Stakeholder value claims

The reviewed studies suggest that enterprises primarily value efficiency improvement, growth opportunities, capability upgrading, and access to external resources through digital platforms (Jeza and Lekhanya, 2022; Lerch and Heimberger, 2022; Xia and Gao, 2018). For many firms, especially SMEs, platforms provide access to markets, knowledge, and complementary assets that would otherwise be difficult to obtain (Gusenbauer et al., 2025; Ibarra-Fretell and Grijalva-Salazar, 2025).

By contrast, customers tend to value service quality, convenience, personalization, fairness, and privacy protection. Their expectations are not limited to product access but also include reliable interaction, lower transaction costs, and protection against the misuse of personal data (Lerch and Heimberger, 2022; Ruutu et al., 2017).

Service providers seek market access, complementarities, and a sustained demand for technical, consulting, and operational support services. Their value claims often depend on continued participation in platform ecosystems and stable collaboration with firms and platform operators (Le et al., 2024; Sassanelli and Terzi, 2022; Xu and Sukpasjaroen, 2024).

Platform operators typically prioritize control, scale, stability, and ecosystem vitality. They seek to maintain platform functionality; attract participation; enforce rules; and preserve strategic influence over transactions, data flows, and governance arrangements (Dossymbek et al., 2025; Hermes et al., 2020; Masuda et al., 2021).

Regulators and public actors tend to emphasize compliance, legitimacy, market fairness, data security, and broader socioeconomic stability. Their concerns extend beyond firm-level performance and include public accountability, competition order, and the social consequences of platform expansion (Chotia et al., 2025; Drewel et al., 2021; Li and Mei, 2024).

Researchers occupy a distinct position in this value structure. They do not participate in platform exchanges in the same way as firms or users, but they shape the field through knowledge production, conceptual framing, and methodological choices that affect how transformation is interpreted and evaluated (Bygstad et al., 2020; Kuan and Lee, 2023).

4.3.2. Recurring conflicts in platform-enabled transformation

However, these stakeholder value claims do not align naturally. Instead, the literature reveals several recurring conflicts that structure the value dimension of the DP-EDT. The first tension arises between enterprises and platform operators. Firms seek access, flexibility, and growth, whereas platform operators seek control, standardization, and strategic stability. This has created recurring disputes over autonomy, dependence, access conditions, and rule-setting power.

The second recurring tension concerns customers and data-driven platform operations. Firms and platform operators rely on data sharing to improve personalization, coordination, and efficiency; however, customers often value privacy, transparency, and fair treatment. As a result, data-driven value creation frequently conflicts with user expectations regarding control over personal information (Ruutu et al., 2017).

A third conflict emerged between platform expansion and regulatory oversight. Platform operators and participating firms often prioritize innovation speed, scale, and market reach, whereas regulators emphasize compliance, competition fairness, accountability, and social stability. This tension is especially visible when platform growth challenges existing regulatory frameworks or intensifies concerns over data governance and market concentration (Chotia et al., 2025; Li and Mei, 2024).

A fourth recurring tension is asymmetry within the ecosystem itself, especially between SMEs and the dominant incumbents. While platforms may lower barriers to participation, they can also reproduce dependence by concentrating visibility, data, and bargaining power on larger actors. In such cases, platform participation generates both opportunities and vulnerability for smaller firms (Gusenbauer et al., 2025; Ibarra-Fretell and Grijalva-Salazar, 2025).

4.3.3. *Coordination mechanisms in platform ecosystems*

The reviewed literature suggests that these conflicts are managed through a set of recurring coordination mechanisms, rather than through permanent resolution. One mechanism is the use of access rules and governance structures that define participation rights, information boundaries, dispute procedures, and decision-making authority. Such arrangements help stabilize interactions while preserving the degree of openness.

The second mechanism is trust-based coordination. In many contexts, formal rules alone are insufficient, particularly when platform ecosystems depend on repeated interactions, collaboration, and knowledge exchange. Trust arrangements therefore complement formal governance by reducing uncertainty and facilitating cooperation among heterogeneous actors.

The third mechanism involves compliance and data governance. Encryption, access control, monitoring rules, and responsibility allocation balance innovation with privacy, security, and legitimacy. These mechanisms are especially important because value creation depends heavily on sensitive or shared data.

The fourth mechanism is incentive alignment. Platforms frequently rely on pricing rules, participation rewards, service differentiation, or partnership arrangements to align the interests of firms, complementors, and users. Effective coordination, therefore, depends not only on rule enforcement but also on whether actors perceive continued participation as beneficial.

Taken together, these mechanisms indicate that the value dimension of DP-EDT is not related simply to stakeholder benefits. It concerns how competing claims are selectively aligned through governance, trust, compliance, and incentives. In this sense, value creation in a platform-enabled transformation is inseparable from conflict management.

Fig. 6 summarizes the stakeholder ecosystem. It should be read as a structured field of interdependence in which value is produced through ongoing negotiations among efficiency, control, privacy, legitimacy, and access. This perspective helps connect the value dimension to the conceptual and methodological dimensions of the dialectical framework because stakeholder conflicts both reflect platform design choices and expose the limits of existing measurement approaches.

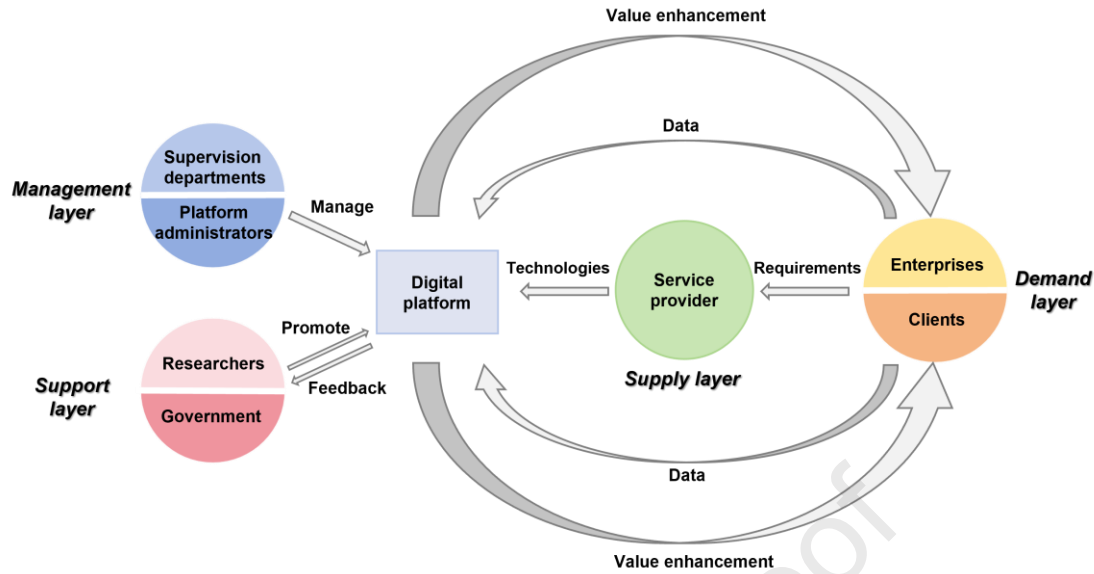


Fig. 6. Stakeholder ecosystem.

5. Discussion

This study developed a dialectical systems framework for understanding DP-EDT based on bibliometric mapping and qualitative content analysis. The framework integrates three dimensions: concept, method, and value. The concept dimension captures the substantive components of platform-enabled transformation, including drivers, platform types, transformation pathways, governance mechanisms, and effects. The method dimension captures how these components are observed, measured, and interpreted in existing research. The value dimension captures how stakeholder interests are created, constrained, redistributed, and coordinated through platform-based transformation. Together, these dimensions explain DP-EDT as a process shaped by technological design, empirical representation, and stakeholder value claims.

The main implication of the framework is that contradictions form a central feature of DP-EDT. The reviewed literature shows recurring tensions within and across the three dimensions. At the conceptual level, platform-enabled transformation involves openness and control, efficiency and resilience, data sharing and privacy, scalability and adaptability, and short-term value capture and long-term ecosystem viability. At the methodological level, these tensions are often translated into measurable variables, models, and indicators. At the value level, they appear as conflicts among firms, platform operators, customers, complementors, regulators, and wider public actors. The dialectical framework connects these patterns and explains how platform governance depends on continuous coordination across levels.

The first cross-dimensional conflict appears between the concept and method dimensions. Conceptually, DP-EDT involves dynamic and multilevel tensions, including openness and control, efficiency and resilience, and data sharing and privacy. Empirically, many studies rely on cross-sectional and variable-based designs that capture stable relationships among platform capabilities, organizational factors, and performance outcomes. This creates a mismatch between conceptual complexity and methodological representation. More integrated research

designs are needed to trace how platform tensions evolve across transformation stages. Mixed, staged, longitudinal, and comparative designs can connect quantitative measurement with process-level explanation and improve understanding of why similar platform configurations generate different outcomes across contexts.

The second conflict appears between the method and value dimensions. Existing measurement approaches often focus on efficiency, performance, adoption, innovation capability, and scale because these outcomes are easier to quantify and compare. The value dimension involves broader concerns, including fairness, privacy, inclusion, legitimacy, accountability, stakeholder welfare, and long-term ecosystem sustainability. This creates a method–value mismatch. Future evaluation frameworks should combine organizational performance indicators with actor-centered evidence and governance-related indicators. Such frameworks can show how value is distributed across platform ecosystems, which actors absorb risk, and how coordination mechanisms influence the balance between economic outcomes and social consequences.

The third conflict appears between the concept and value dimensions. Many platform designs and transformation pathways prioritize scalability, efficiency, integration, and value capture. These priorities can intensify concerns over privacy protection, equitable participation, regulatory legitimacy, dependence, and ecosystem sustainability. Platform architectures often embed strategic preferences that allocate control, visibility, data access, and bargaining power unevenly among actors. Coordination mechanisms such as selective openness, modular governance, rule-based access, trust arrangements, compliance architecture, and data governance can reduce these conflicts. Their effectiveness depends on how well participation, control, accountability, and adaptability are balanced across transformation stages.

These three conflicts are interconnected. Weak operationalization of conceptual tensions can produce incomplete empirical evidence. Incomplete evidence can privilege easily measurable outcomes and obscure value conflicts. Unresolved value conflicts can also reshape conceptual categories by changing which transformation outcomes are considered important. The framework therefore treats concept, method, and value as interdependent dimensions of the same transformation system. Three coordination paths follow from this interdependence. Concept–method coordination translates platform tensions into empirically traceable research designs. Method–value coordination expands evaluation systems to reflect diverse stakeholder claims. Three-dimensional iterative coordination adjusts conceptual categories, methodological tools, and value targets in response to changing platform conditions.

The findings also show that effective platform governance depends on dynamic recalibration. As platform ecosystems evolve, firms, platform operators, and regulators need to adjust access rules, control structures, data governance arrangements, and incentive systems. Transformation outcomes are shaped by how conflicts are coordinated across stages of platform development. The dialectical systems framework adds value by explaining which elements matter in DP-EDT, how tensions among these elements emerge, and how coordination mechanisms shape transformation results.

The conclusions of this review should be read in light of the limitations of the included evidence. First, a substantial share of empirical studies uses cross-sectional survey designs,

which limits evidence on how platform governance tensions evolve over time. Second, many studies measure transformation through firm-level outcomes such as performance, adoption, innovation capability, and efficiency, with less attention to distributional effects, regulatory consequences, privacy concerns, and stakeholder welfare. Third, sectoral and regional coverage remains uneven. Manufacturing, SMEs, e-commerce, and platform-based service contexts are more visible than highly regulated or data-sensitive sectors. Fourth, longitudinal, comparative, and mixed-method research remains limited, which constrains the ability to examine stage-dependent transformation mechanisms. These limitations affect the completeness, transferability, and interpretive strength of the review findings.

The findings have policy implications for platform governance. Policymakers and regulators should treat platform-enabled enterprise transformation as a governance process with technical, organizational, and social consequences. In data-sensitive and regulation-intensive sectors, regulatory frameworks should specify rules for data access, portability, interoperability, auditability, privacy protection, and responsibility allocation among platform operators, firms, service providers, and users. Oversight mechanisms should also address power asymmetry within platform ecosystems, especially where SMEs depend on dominant platform operators for visibility, transactions, and data access. Sector-specific regulatory design is needed to balance innovation, competition, data protection, and long-term ecosystem viability.

Fig. 7 illustrates the dialectical systems framework. It summarizes the multidimensional structure of DP-EDT and shows how concepts, methods, and values are linked through conflict and coordination. The figure should be read as a process model of platform-enabled transformation, showing how recurring tensions, governance responses, and stakeholder value claims interact over time.

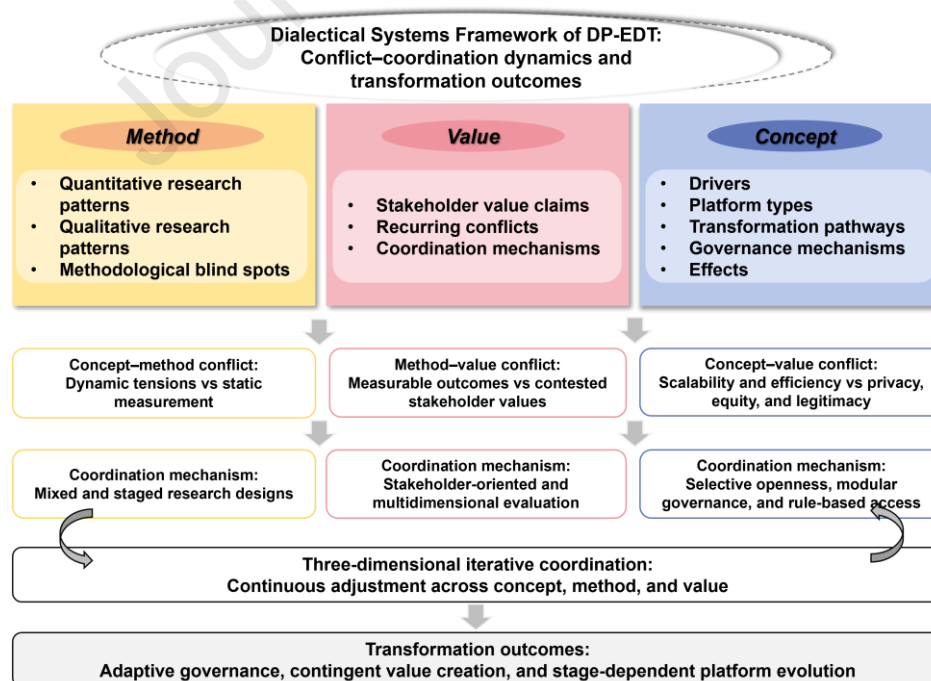


Fig. 7. Dialectical systems framework of digital platform-enabled enterprise digital transformation.

The dialectical framework developed in this study suggests that future research should move beyond the broad calls for wider industry coverage or more mixed methods. Instead, future work should focus on unresolved tensions, coordination mechanisms, and stage-dependent outcomes revealed by the concept–method–value system. Based on the reviewed evidence, four research propositions are particularly important.

Proposition 1: Stage-dependent openness–control tensions deserve closer examination.

This proposition follows from the finding that platform-enabled transformation does not occur through a single governance logic. As firms move from early digital entry to platform scaling and ecosystem orchestration, the balance between openness and control also shifts. Early-stage transformation often benefits from openness, experimentation, and access to external knowledge, whereas later-stage transformation requires stronger rule-setting, coordination, and selective control. Existing studies acknowledge openness–control tensions but usually treat them as static governance dilemmas rather than stage-dependent transformation dynamics. Future research should therefore examine how openness and control are recalibrated across transformation stages and what governance mechanisms, such as modular architecture, layered access, or selective participation rules, enable firms to maintain participation without losing strategic coherence.

Proposition 2: Method–value mismatch remains a major obstacle in evaluating stakeholder trade-offs.

This proposition emerges from the finding that dominant research designs privilege measurable outcomes, such as efficiency, performance, and adoption, while paying less attention to value conflicts involving fairness, privacy, legitimacy, inclusion, and long-term stakeholder welfare. Existing studies often identify whether digital platforms create value, but provide weaker evidence on how that value is distributed and whose interests are prioritized. Future research should address this mismatch by developing stakeholder-oriented evaluation frameworks that combine organizational performance indicators with actor-centered evidence. Longitudinal, comparative, and mixed-method designs are especially useful for examining how value trade-offs change over time and how governance arrangements influence the distribution of benefits and risks.

Proposition 3: Governance conflicts in data-sensitive and regulation-intensive sectors require context-specific analysis.

This proposition follows from the reviewed evidence that tensions between innovation, data sharing, compliance, and accountability become especially pronounced in sectors characterized by high data sensitivity, strong regulation, or complex public consequences. In these contexts, platform-led transformation is shaped not only by technical capability but also by the interaction among platform operators, firms, regulators, and users. Although current studies recognize the importance of data governance and compliance, they often do so at a general level and rarely explain which governance designs are most effective under different institutional conditions. Therefore, future research should investigate how rule-based access, compliance architectures, auditability, trust mechanisms, and incentive alignment operate in sectors such as healthcare, energy, finance, and industrial platforms.

Proposition 4: Transformation conflicts vary across start-ups, incumbents, and traditional

sectors, but remain underexplored.

This proposition is derived from processual analysis of transformation pathways. Start-ups often use digital platforms for experimentation, market access, and rapid scaling. Incumbent firms are more likely to emphasize process integration, business model renewal, and organizational reconfiguration. In contrast, traditional sectors face stronger path dependence, legacy constraints, and coordination friction. Existing studies frequently discuss platform-enabled transformation in generic terms without sufficiently distinguishing how tensions differ across organizational types. Future research should compare these contexts more explicitly and examine how transformation conflicts—such as flexibility versus standardization, autonomy versus dependence, and innovation versus legacy compatibility—are shaped by firm history, resource endowment, and ecosystem position.

Taken together, these propositions suggest that future research should not simply ask whether digital platforms support enterprise digital transformation but under what conditions, through which coordination mechanisms, and for whose benefit. This would allow the field to move from a broad descriptive synthesis toward a more processual, conflict-sensitive, and theoretically distinctive research agenda.

6. Conclusions

This review explains how DP-EDT can be understood from a dialectical system perspective. In response to RQ1, this review identifies the core components of DP-EDT—drivers, platform types, transformation pathways, governance mechanisms, and effects—and shows that these elements are interdependent rather than isolated. In response to RQ2, this review demonstrates that these components interact through recurring tensions and coordination processes across the concept, method, and value dimensions. In response to RQ3, this review develops a dialectical systems framework that explains how cross-level transformation processes are shaped by the interaction between platform design, empirical representation, and stakeholder value claims.

The main theoretical contribution of this study lies in moving the literature beyond the fragmented treatment of digital platforms, organizational change, and ecosystem governance. The framework connects these strands by showing how transformation unfolds through tension–coordination dynamics. More specifically, it integrates conceptual structures, methodological patterns, and value conflicts into a single analytical system and clarifies why transformation outcomes depend not only on platform adoption but also on how competing demands are negotiated across actors and stages.

This review also has several practical implications. For managers, platform-enabled transformation should be governed as an adaptive process rather than a one-time technological upgrade. For platform operators and policymakers, the review shows that effective transformation depends on balancing openness and control, data sharing and protection, and short-term efficiency with long-term ecosystem viability. The framework provides a structured basis for thinking about platform governance, coordination design, and value allocation in enterprise transformation.

This study has several limitations. First, the review focused on peer-reviewed English-language journal articles and book chapters indexed in Web of Science and Scopus. This choice

supported cross-disciplinary coverage and bibliometric mapping, while also excluding some conference proceedings, practitioner reports, grey literature, non-English publications, and studies indexed only in specialist computer science or business databases. Second, the search strategy centered on platform-related terms. Related information systems studies using terms such as digital capability, IT capability, information systems, or IT-enabled transformation may therefore be underrepresented. Third, the dataset captures management-oriented and ecosystem governance perspectives more strongly than infrastructure-oriented and technical implementation perspectives. Future updates or companion reviews could extend the search to major information systems and computer science databases, conference proceedings, and non-English sources, and compare their findings with the management-oriented synthesis reported in this review.

Overall, this review shows that DP-EDT is an evolving governance system shaped by the interaction among platform architectures, research methods, and stakeholder values. Platform-enabled transformation develops through recurring adjustments among technological design, empirical representation, and stakeholder value claims. These interactions shape the trajectories and outcomes of enterprise digital transformation in platform ecosystems.

Declaration of competing interests

The authors declare that there are no conflicts of interest.

Data availability

The final list of included studies, data extraction framework, coding scheme, and bibliometric configuration files will be deposited in the OSF project repository. The repository will include the included-study list, coding categories, and VOSviewer configuration where feasible. Full-text articles cannot be shared because of copyright restrictions. Full-text exclusion reasons are reported by category in the PRISMA flow diagram and Methods section. Additional coding records are available from the corresponding author upon reasonable request.

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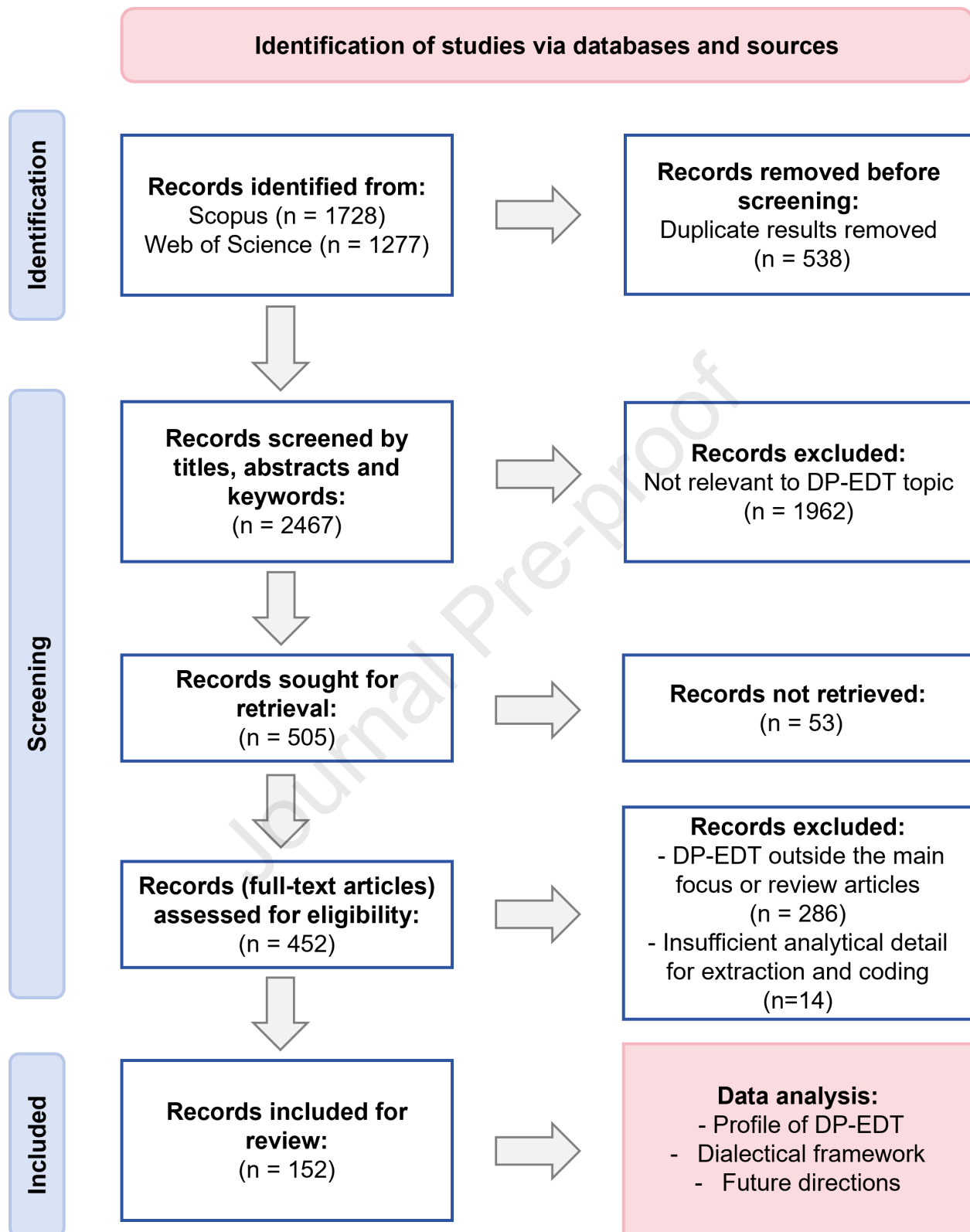
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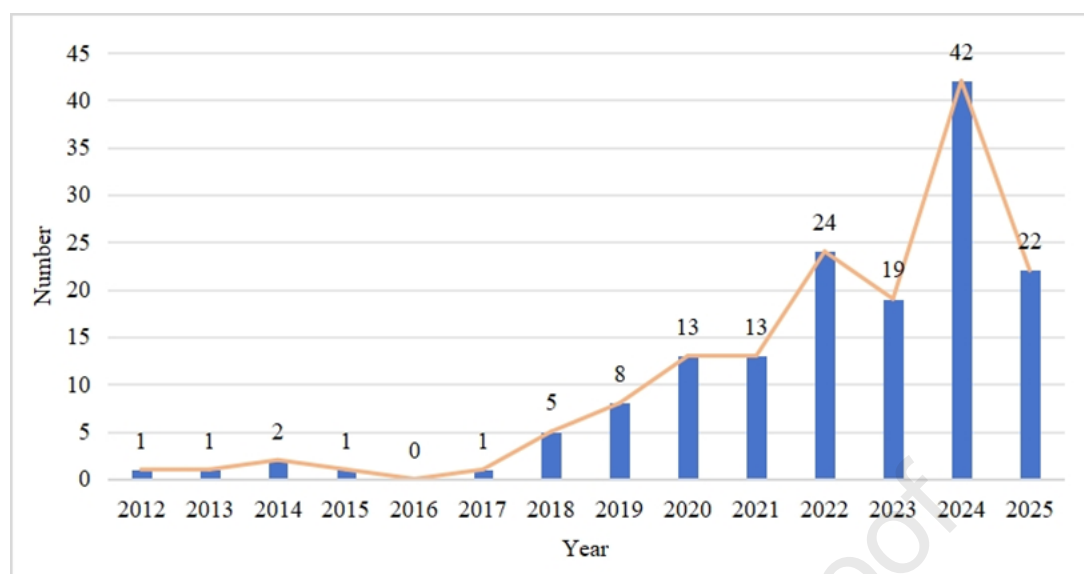
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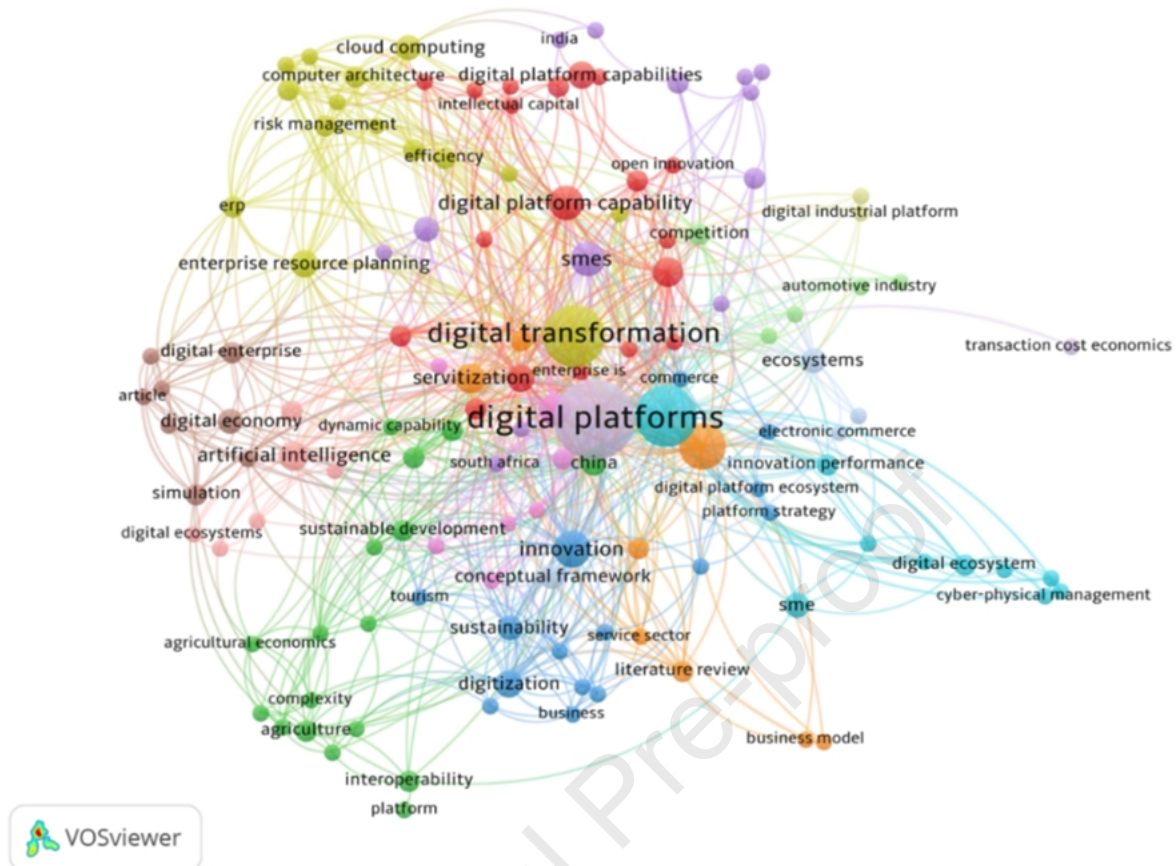
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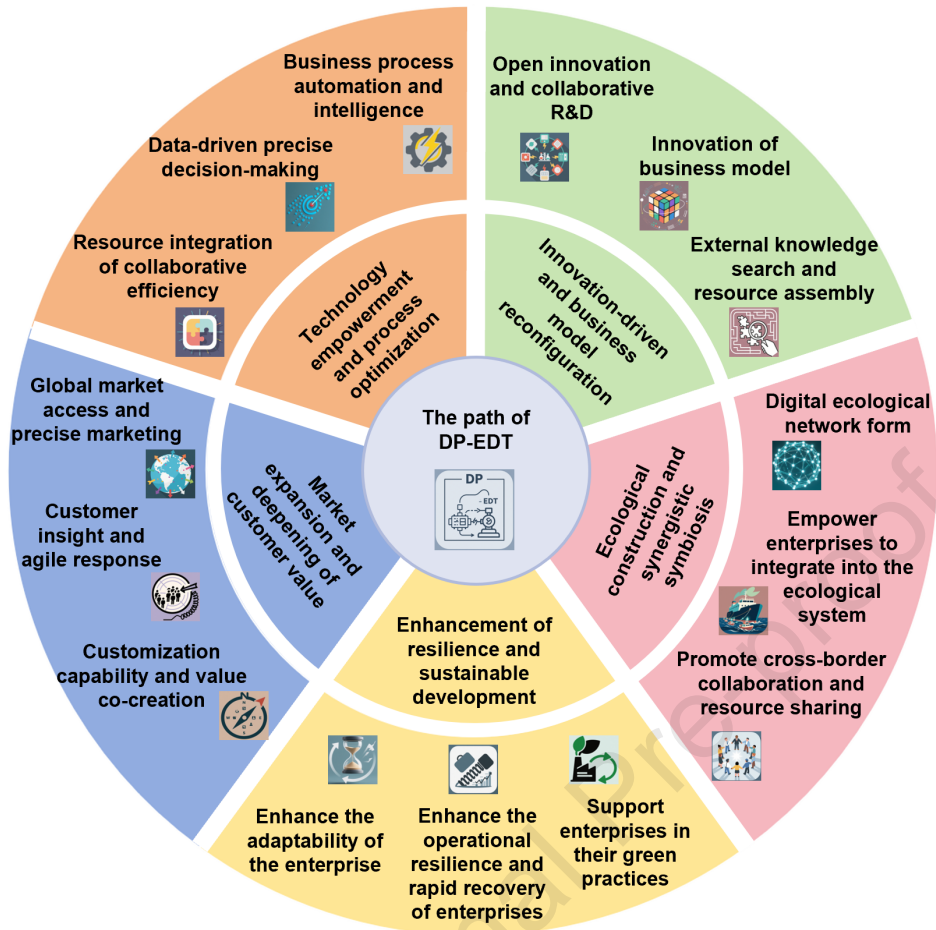
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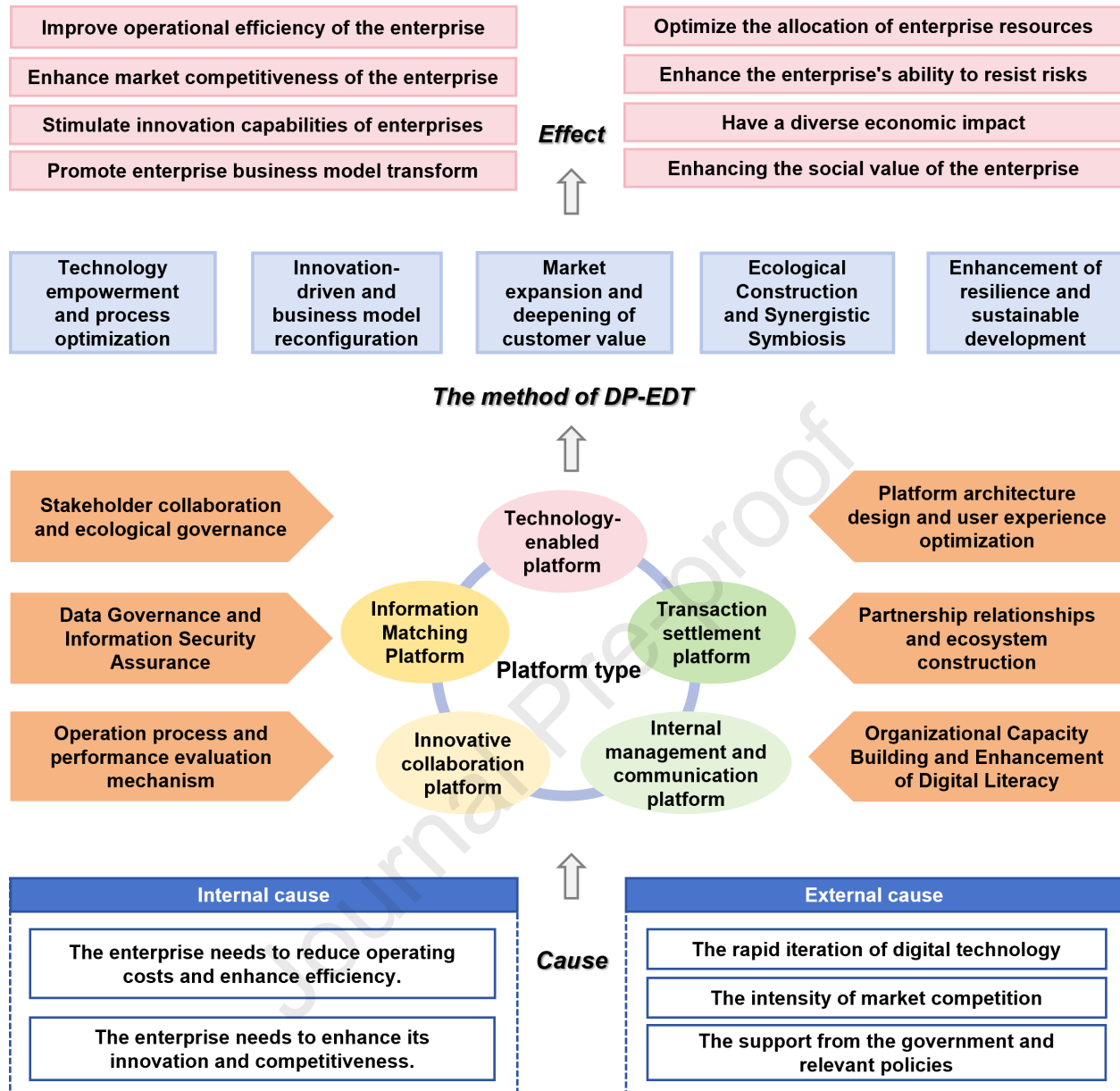
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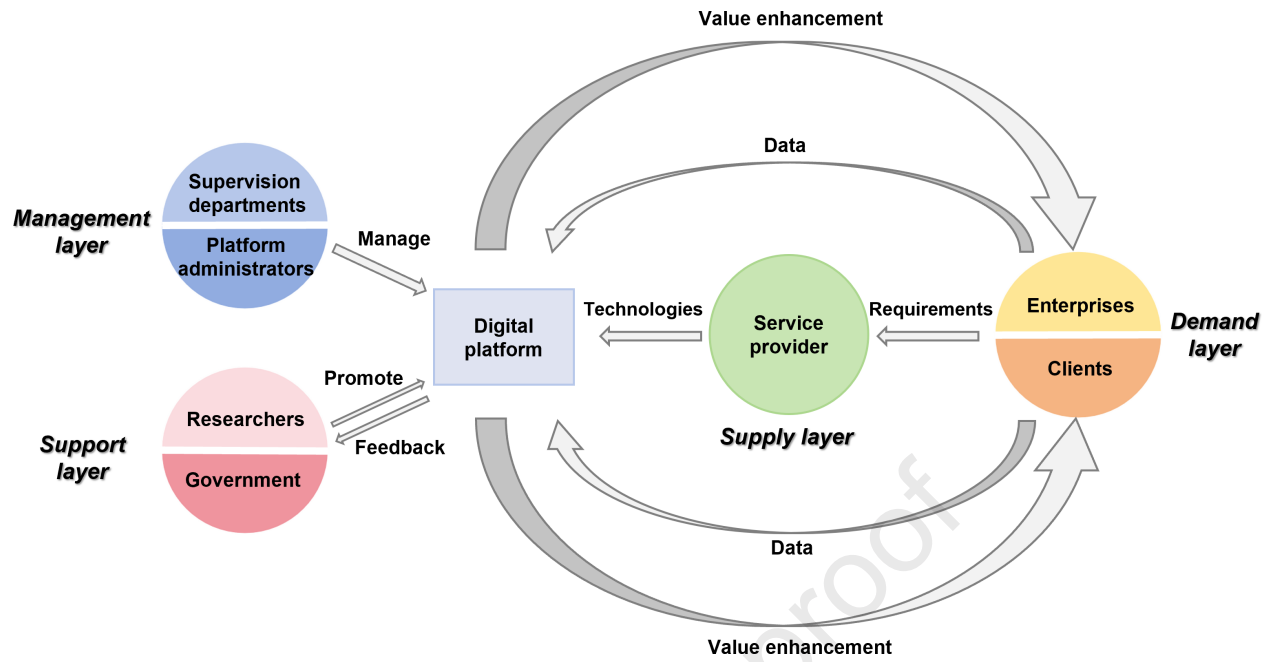


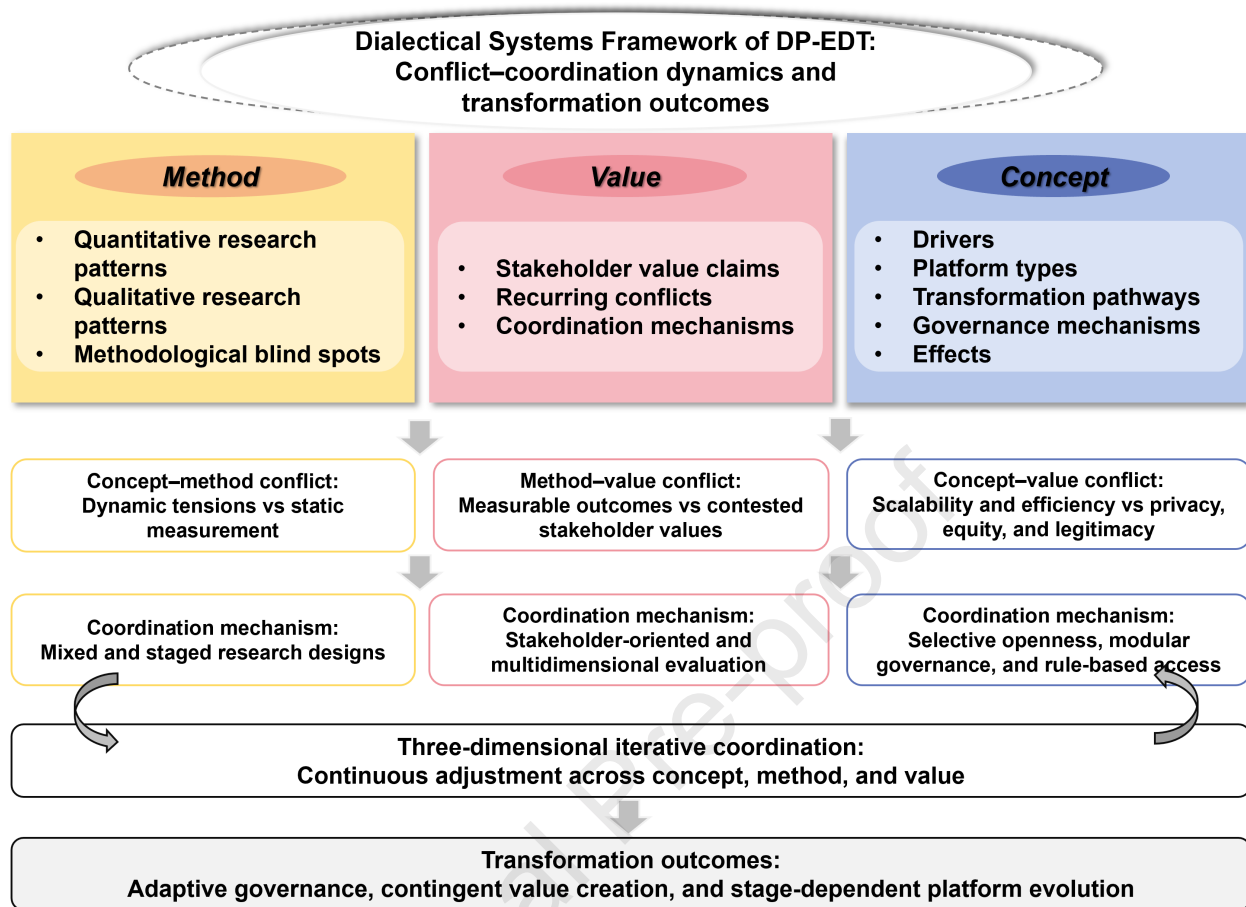












Digital Platform Ecosystems and Enterprise Digital Transformation: A Dialectical Systems Framework for Sustainability

Highlights

- Proposes a novel dialectical systems framework to guide enterprise digital transformation through digital platform ecosystems.
- Integrates three core dimensions—concepts, methods, and values—to provide a comprehensive analytical lens on platform-enabled transformation.
- Explores internal and external mechanisms of conflict and coordination within and across these dimensions.
- Employs a systematic combination of bibliometric mapping and qualitative content analysis to review and synthesize the existing literature.
- Identifies eight forward-looking research directions that inform future academic work and support evidence-based policy design and managerial decision-making.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The author *Click here to enter your name* is Choose an item for *Click here to enter the journal's name* and was not involved in the editorial review or the decision to publish this article.

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The authors declare that there are no conflicts of interest.